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Review of wildfire modeling considering effects on land surfaces

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ABSTRACT

Wildfires are part of the natural cycle of life in vegetated regions. The apparent increase in size and frequency of recent years reflects land management legacy, expansion of human activity, and changes in climatic conditions. The primary objectives of the review are: (1) to introduce hydrologists, soil scientists and ecologists to principles and advances in modeling of wildfire dynamics and rates of spread to improve understanding of capabilities and limitations offered by modern wildfire models; (2) to highlight persistent omissions of wildfire effects on soil processes and how soil and hydrology communities can harness wildfire models to bridge these gaps and provide the necessary boundary conditions for quantifying thermal alterations of soils. The review aims to enhance cross-disciplinary understanding critical for advancing quantitative representation of wildfire impacts on soil and hydrologic processes and formulating boundary conditions for subsequent hydro-ecological recovery of fire-affected landscapes. The review presents key elements used to represent wildfire characteristics, processes and metrics considered in wildfire science. We then provide an overview of wildfire modeling approaches and models used for research and for prescribed burning planning that include wildfire behavior models, semi-physical and mechanistic large eddy simulation models. We highlight key features and assumptions, and common applications of each class of models and their inputs (spatially resolved databases of fuel maps, digital elevation, land cover, and weather). The second section of the review proposes a path towards building a systematic framework for quantifying soil, ecological and hydrological wildfire-induced alterations to parameterize post-fire hydrologic responses and predict landscape recovery rates across seasonal to climatic time scales.

1. Introduction

Wildfires play an important role in the ecology of vegetated landscapes, causing natural disturbances that have always been an important component of vegetation life cycle by affecting landcover and function at climatic and evolutionary time scales (Mclauchlan et al., 2020). Rapid human expansion into fire-prone regions coupled with recent climatic trends that promote the onset and spread of wildfires (Abatzoglou and Williams, 2016) provide the impetus for better understanding and predicting wildfire behavior. Traditionally, wildfire sciences and modeling have focused on fire safety engineering, wildfire control strategies, and smoke and particle emissions to the atmosphere. The primary motivation of this review of wildfire physical processes and models was the need for a quantitative (and ultimately predictive) framework for assessing impacts of wildfires on soil and landscape functions and hydrologic process. Quantitative predictions and assessment of wildfire effects are important for formulating land (and fuel) management practices to mitigate adverse impacts of large wildfires and to develop restoration strategies for fire-affected landscapes. Advances in wildfire models with availability of high-performance computing and direct links with weather models and an array of observational capabilities offer new insights at unprecedented level of detail into this natural (and prescribed) phenomenon. Nevertheless, considerable gaps remain concerning spatial patterns and fire-fuel-atmosphere interactions that affect fire intensity and spread behavior and impact landscape ecology and hydrology. Specifically, feedback between wildfire characteristics and soil and hydrologic impacts are virtually absent in present wildfire modeling frameworks. This review focuses on identifying the origins of

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this representation gap within wildfire models and proposes a path forward for linking wildfire models with thermal alterations of soils that affect ecohydrological functions of fire-affected landscapes.

Using the US National Forest lands as a benchmark, Kenward et al. (2016) found that wildfire activity and spatial extent have been accelerating over the past 45 years with more than six times the area being burned annually in the 2010's relative to the 1970's. The length of the wildfire season has grown by 105 days during the same period to the point that defining a fire season is becoming meaningless with fires occurring nearly year-round. In addition to increasing wildfire activity and impact, the expansion of human activity with has led to more people are living at the wildland-urban interface (WUI) thus putting themselves and property at increased risk of damage by wildfires (Kenward et al., 2016). The steady expansion of the WUI in the US, Canada and Australia coupled with the legacy of fire exclusion or suppression during the past century have not only changed the perception of wildfires as natural phenomenon (and associated risks) but have also altered the fire regimes themselves (Hagmann et al., 2021) with more frequent, larger and more severe wildfire events. Recent studies show that the sizes of wildfire-affected areas have been steadily increasing in the past few decades as illustrated in Fig. 1 from Abolafia-Rosenzweig et al. (2022) focusing on burned areas in the western US.

In part, changes in wildfire regimes have been attributed to human-induced landscape management that resulted in changes in the vegetation landscape and fuel loads (Hagmann et al., 2021; Prichard et al., 2021), other authors have highlighted persistent changes in climatic conditions with drier and warmer conditions in the western US (Abatzoglou et al. 2016, Abolafia-Rosenzweig et al., 2022), the reduction in natural conditions that suppress and extinguish wildfires (such as nighttime humidity, see Balch et al., 2022), and increased human activity with more frequent ignitions (Abatzoglou et al., 2018; Li and Banerjee, 2021).

Mounting evidence suggests that increased frequency and size of wildfires are due to a combination of factors mentioned above (primarily changing climate and anthropogenic land management, Abatzoglou and Williams, 2016) as also seen from Earth observation platforms and recent occurrences of very large wildfires in Australia, Siberia, Canada, and other regions of the world. For example, the total burned area from 2019 to 2021 in Australia and Siberia alone exceeds 550,000 km^2 (135.9 M acres); to place these numbers in proportion with recent extensively reported fires in the media, the 2021 Dixie fire combined with the 2020 Bay Area composite fire (largest on record in California) have affected "only" 7320 km^2 (1.81 M acres). Clearly, wildfire impacts on society and infrastructure cannot be measured solely in terms of burned area; Doerr and Santín (2016) analyzed the western-biased perception of wildfires and the need to facilitate a broader and more informed understanding of wildfire realities in a changing world. A recent global review of wildfires from 2001 to 2019 (Jones et al., 2022) concluded that while burned area has reduced globally by 27% over the past two decades, certain regions in the world such as the Northwestern US have experienced a 50% increase in the burned area attributed primarily to a warming climate. Tyukavina et al. (2022) reported that forest fires result in 3 million hectares of tree cover loss per year compared to 2001 (>25% of tree loss over the past two decades). Nearly 70% of the global tree cover loss due to forest fire occurs in the boreal regions (i.e., Siberia, Canada, Alaska) due to the rapid warming up of these regions and limited intervention due to the regions' remoteness. The recent global reviews of Tyukavina et al. (2022) and Jones et al. (2022) provide a consistent picture of less burned areas in the African Savanna due to agriculture and human control that is countered by expansion of Boreal Forest fires due to conducive (warmer and dry) climatic conditions. An important explanatory element related to wildfire-urban interface invokes the legacy of land management during the past century in North America and Australia (fire suppression, grazing) that modified the vegetation and fuel landscape thus introducing increased vulnerability for more intense and larger wildfires (Hagmann et al., 2021; Prichard et al., 2021).

A warmer and drier climate coupled with a legacy fuel buildup contribute to wildfire intensification (Abatzoglou et al., 2016; Jones et al., 2022) with increasing frequency and size of severe wildfires that affect not only vegetation cover, but directly impact soil resources and alter the hydrology of burned regions (Williams et al., 2022; Ma et al., 2020; Ebel et al., 2023). To better quantify the complex interactions between wildfire type, intensity, duration, fuel structure and other metrics, we seek to formulate quantitative links between fire physical characteristics and expected signatures on soil structure and hydrologic response that may last many years after the wildfire event. Alexander (1982) pointed out that wildfires are not binary events (i.e., burned vs not burned), they involve spatially and temporally variable processes whose characteristics and patterns determine different soil and hydrologic outcomes. For example, harnessing information on fire line radiative energy density (i.e., the amount of heat released during passage of flaming front – to be elaborated shortly) is important beyond impacts on tree damage and mortality (van Wagner, 1967) or as a comparative metric across different wildfires (Alexander, 1982). The translation of such information available from wildfire models, remote sensing, and from post-fire assessment can assist with quantifying thermal alterations of soil properties towards predicting a range of hydrologic and ecological post-fire responses and recovery rates.

The primary objectives of this review are: (1) to introduce the

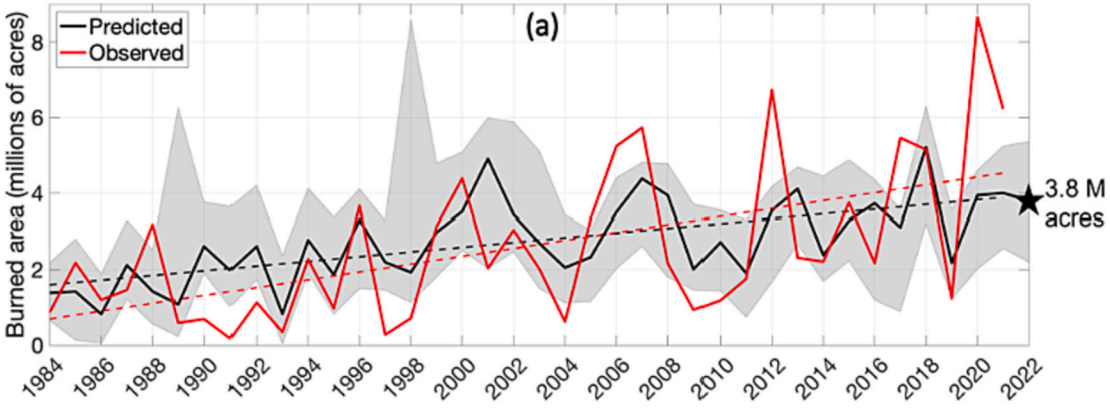


Fig. 1. A time series of observed annual summer fire season burned areas in the **western US** showing an increasing trend since 1984. The predicted burned area for 2022 is 3.8 million acres (black star), which is the eighth largest in the model's 39-year record. Gray shading shows the 95% ensemble range. Black and red dashed lines show linearized trends of modeled and observed burned area from 1984 to 2021 (from Abolafia-Rosenzweig et al., 2022). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

hydrology, soil, and earth science communities to key principles that underlie physical processes and modeling approaches used for describing wildfire behavior; (2) to provide insights that can facilitate better and more nuanced assessment of likely impacts of wildfires on soil and hydrologic processes; and (3) to identify modeling gaps with respect to wildfire effects on soil and how to bridge these gaps. Specifically, we will review key developments in quantitative representation of wildfire characteristics and the primary processes and metrics commonly considered in wildfire science. We will then present the primary models presently used for research and prescribed burning including wildfire behavior models, semi-physical wildfire spreading models and mechanistic large eddy simulation-based models. We will highlight key features, assumptions, and applications for each group of models. We will briefly discuss recent enabling developments in data science with availability of spatially resolved and regularly updated databases of fuel maps, digital elevation, land cover, and climate. Finally, we will provide an outlook regarding how these developments and modeling capabilities can be used to better assess impacts on soil structural and hydrologic processes considering post fire impacts on hydrological processes at the soil profile to catchment scales.

2. Wildfire models

Some of the earliest efforts to quantify and simulate wildfire spread were reported by Fons (1946), Molchanov (1957), Byram (1959), Emmons (1964), van Wagner (1962, 1967), Hottel et al. (1965) and McArthur (1967). These studies applied semi-physical concepts and models of increasing complexity that led to the formulation of a widely used fire spread model of Anderson (1969), Frandsen (1971) and Rothermel (1972). The Rothermel model considers a one dimensional steadily and isothermally spreading inclined fire line over a flat surface overlain by a uniform fuel bed. The fuel bed is characterized by its moisture content, volumetric fuel density and packing coefficients and fuel element size expressed in terms of volume to surface area ratio (of wood elements). Rothermel represented the rate of fire line spread R (m/min; in some models it is referred to as *ROS*) in the following closed-form expression:

$$R = \frac{I_R \xi (1 + \phi_w + \phi_s)}{\rho_b \epsilon\, Q_{ig}} \tag{1}$$

where:

Table 1
Classes of variables for Rothermel's fire spread model with most closely related parameters in Eq. 1 (modified from Perry, 1998).

Fuel particle properties		Fuel geometry		Environmental parameters	
Heat content	I_R	Fuel load	ρ_b	Wind speed	ϕ_w
Mineral content	I_R	Particle size	ε (1→0); ξ; I_R	Fuel moisture	Q_{ig}
Particle density	I_R	Fuel bed depth	ρ_b	Slope steepness	ϕ_s

Table 2
Intrinsic thermochemical properties for Ponderosa Pine and Western White Pine dead foliage (From Albini, 1981 and references therein).

Property	Units	Ponderosa Pine	Western White Pine
Heat required to raise unit of dry mass from 25 to 325^0 C	kJ/kg	603	603
Heat required to raise unit of dry mass from 25 to 500^0 C	kJ/kg	1017	1017
Fraction of initial dry mass converted to char at 500^0 C	(−)	0.2717	0.3370
Fraction of initial dry mass that is mineral ash	(−)	0.0405	0.0424
Heat of combustion of the ash free fuel	MJ/kg	21.55	20.54

Table 3
Different wildfire model classification schemes: (a) focusing on wildfire behavior and spread (Weber, 1991; Perry, 1998); (b) emphasis on process coupling and scales (Bakhshaii and Johnson, 2019); and key features of contemporary fully physical models (Morvan, 2011).

(a) General wildfire model classification modified from Perry (1998)

Type of model	Authors	Components/Applications
Physical- mechanistic	Albini, 1986	Theoretical and experimental - *radiative transfer reradiation and convection*
	Fons, 1946	Theoretical and experimental - *radiative transfer*
	Frandsen, 1971Grishin, 1997	Theoretical - *energy flux conservation*
Semi-physical		Theoretical - *heat flux and chemical kinetics*
	Rothermel, 1972	Theoretical - *energy flux conservation*. Core of various fire behavior models (e.g., BEHAVE)
	McArthur, 1966	Australian grassland fuels (fire shape model)

(b) Model classification based on process coupling and scales (Bakhshaii and Johnson, 2019)

Coupled model	Coupling$^+$	Spatial scale	Domain	Time scale	CPU usage
FIRETEC	One way	0.001–1 m	1 km^2	1 s – 2 h	No real time
WFDS	One way	0.001–10 m	> 2 km^2	1 s – 4 h	No real time
CWFE	Two ways	10–30 m	> 10 km^2	1 s - hrs	No real time
WRF-Fire/ SFIRE	Two ways	10 m – 10 km	1–100 km	1 s - days	real time

(c) Classification of several fully physical wildfire models modified from Morvan (2011)

Model	Numerical	Low Mach	Turbulence	Combustion model	Multi fuel	Low scale	Large scale
FIRETEC	2D explicit	No	LES	No	Yes	No	Yes
WFDS	3D explicit	Yes	LES	Yes	Yes	Yes	Yes
FIRESTAR	2D implicit	Yes	TRANS	Yes	Yes	Yes	Yes

+ two-ways coupling implies that the wildfire model represents processes that alter local atmospheric/weather processes (i.e., the fire generates its own weather).

R = the forward rate of spread of the flaming front (m/min)
I_R = the reaction intensity (kJ/m^2/min)
ξ = the propagation flux ratio (dimensionless)
ϕ_w = wind coefficient (dimensionless)
ϕ_s = slope factor (dimensionless)
ρ_b = fuel bulk density (kg/m^3)
ε = effective heating number (dimensionless)
Q_{ig} = the heat of pre-ignition of the fuel (kJ/kg)

The ingredients and key steps involved in model development were clearly and concisely described in Rothermel's (1972) report highlighting salient physical features of fire line behavior, fuel characteristics, key assumptions, and empirical approximations required for closed-form representation of fire line spreading for a wide range of conditions (Fig. 3a). A recent review by Andrews (2018) reexamines subsequent developments spawned from Rothermel's model. For clarity, Table 1 (modified from Perry, 1998) presents the different classes of input variables for application of the Rothermel model and key parameters in Eq. 1 used for representing rate of fire spread. Tables 1-3 in Andrews (2018) provide an extensive review of parameters and equations used for practical applications and Fig. 2 below depicts a flow chart

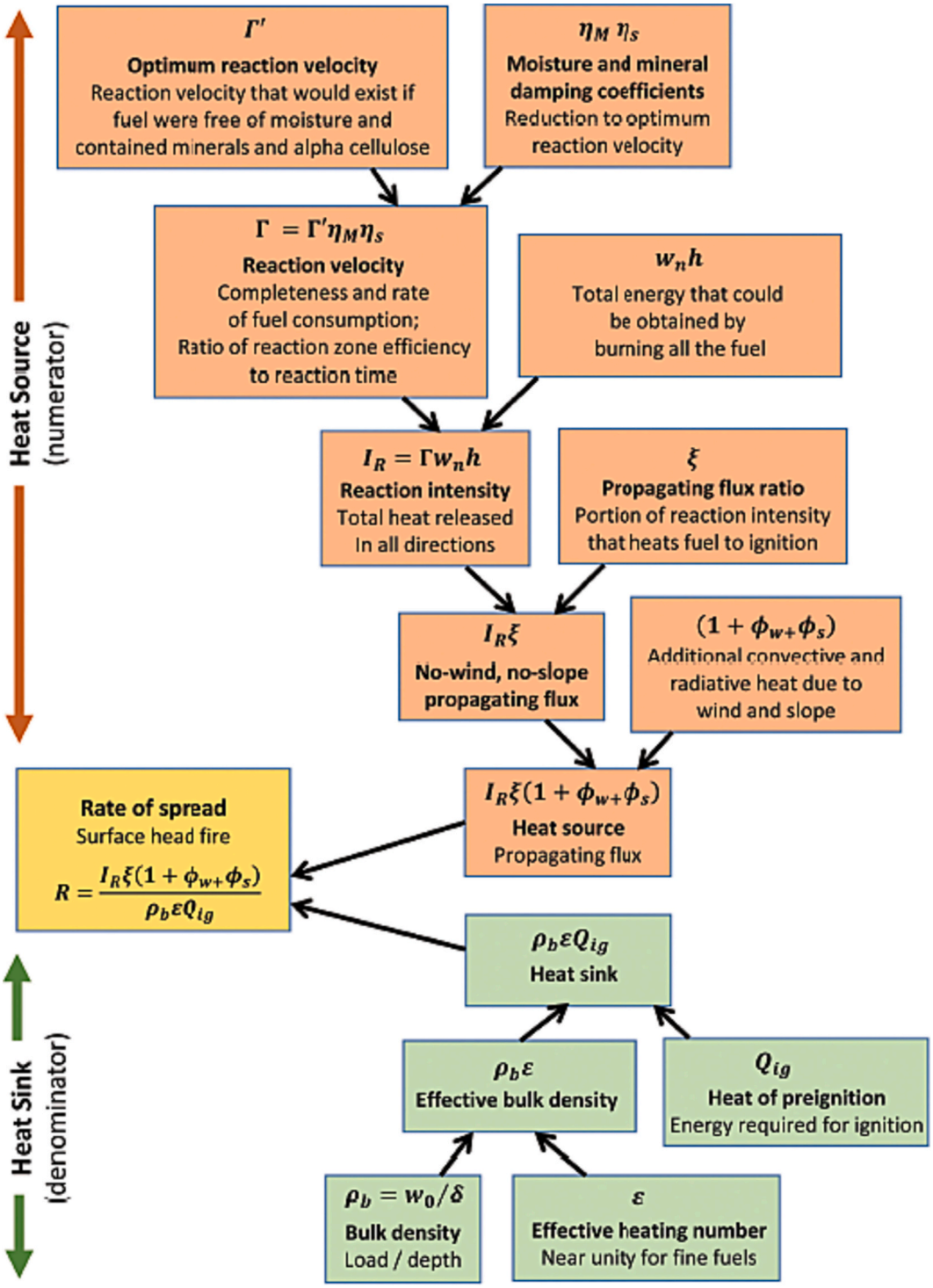


Fig. 2. Flow of calculations in the basic Rothermel (1972) fire spread model for one size class of fuel (from Andrews, 2018).

of the calculations of Rothermel (1972) rate of fire spread. Many of the variables have been constrained using experimental results (Rothermel and Anderson, 1966; Andrews, 2018).

In a series of influential studies, Albini (1981, 1985, 1986) introduced important physical ingredients that greatly enhanced modeling capabilities to predict fire spreading dynamics by considering details of energy exchange at the fire line, interactions with wind and reaction rates, convection and radiation and fire spotting (firebrands). In Albini (1981) the one-dimensional model considers a mixing-controlled reaction rate for the one-step conversion of fuel and air to combustion products that subsequently help place empirical expressions for flame length, height, and angle on a more physical basis. The model also clarified reliance of thermochemical aspects of burning on fuel moisture and mineral content and other properties (see Table 2 adapted from Albini, 1981).

An important result derived by Albini (1981) is the relations between flame height h_F, mean horizontal wind speed U, and the flame tilt angle α. This is a crucial geometrical variable for flame scorching height and for radiative heating projected ahead of the fire line.:

$$tan\ \alpha = \frac{U}{\sqrt{2gh_F/3}} \tag{2}$$

Both the flame height h_F and the flame tilt angle α are required for calculating view factors (Mudan, 1987) for estimating the fraction of radiative energy intercepted on surfaces exposed to the flame (see Fig. 4 for illustration). Emmons (1964) and Hottel et al. (1965) are credited with systematic consideration of the role of radiation in heating up the fuel bed ahead of a flame of height h_F. These concepts were consolidated in the work of Albini (1985) in which the flame's radiant energy, the shape of the flame and the motion of a two-dimensional ignition interface were mathematically expressed. Morvan (2011) have analyzed the length scales involved in wildfire behavior: (i) the turbulent integral length scale I_T for the atmospheric boundary layer governing wind-flame interactions; and (ii) the radiative extinction length δ_R (characterizing radiative transfer through a vegetation layer). While I_T reflects the effects of canopy height (H_{Fuel}) on the atmospheric boundary layer (differently in the horizontal and vertical directions) as:

$$\begin{aligned} &I_T \backsimeq H_{Fuel}\ horizontally \\ &I_T \backsimeq H_{Fuel}/3\ \ vertically \end{aligned} \tag{3}$$

The radiative extinction length is related to fuel structure, specifically the surface area to volume ratio, σ_S, and the packing ratio a_S (volume fraction of solid vegetation). We note that both parameters feature prominently in Rothermel's (1972) fire rate of advance model (and in other physical fire models). Morvan (2011) expressed the radiative extinction length δ_R as:

$$\delta_R = \frac{4}{a_S\sigma_S} \backsimeq \frac{2{\cdot}H_{Fuel}}{LAI} \tag{4}$$

where LAI is the vegetation leaf area index (total area of leaves over the projected area of tree canopy). Typical values for δ_R (m) for different fuels: 4.75 (Boreal Forest), 0.25 (Mediterranean pine forest); 0.15 (shrubs); 0.1–0.5 (grass) and 0.025 for fuel bed. The radiative heat intensity field I_{Rad} (W m^{-2} str^{-1}) is expressed by the following radiative transfer equation:

$$\frac{dI_{Rad}}{ds} = \sigma_a\left(\frac{\sigma T^4}{\pi} - I_{Rad}\right) + \frac{1}{\alpha_G}\sum_k\left[\frac{\alpha_S\sigma_S}{4}\left(\frac{\sigma T_S^4}{\pi} - I_{Rad}\right)\right]_k \tag{5}$$

With the temperature of the gas T and the solid fuel particles T_S, the gas volume fraction a_G, the gas and soot extinction coefficient σ_a and σ is the Stefan–Boltzmann constant (5.67 × 10^{-8} W m^{-2} K^{-4}). The main point of Eq. (5) is to highlight the direct link between the temperatures of the gases (1st bracketed term on the right-hand side,) and that of solid particles (2nd term on the right-hand side), summed over k different fuel size and type classes) on the radiative energy of emitted by the fire line. Albini (1996) presented an iterative solution to the radiation transfer equation. Since the temperatures at the fire line fluctuate significantly with turbulence, details of averaging of the nonlinear terms (T^4) are important for fire spread predictions and remain a challenge for representation of turbulence-radiation interactions (Morvan, 2011).

Summarizing, while the notion that radiation dominates heating and drying of fuel ahead of a fire line and prepares the fuel bed for ignition remains widely accepted, the actual process of ignition seems to be dominated by physical flame contact of the thermally primed fuel as recent studies by Finney et al. (2015) have shown. In their experiments, Finney et al. (2015) show that wildfire spread is governed by tight coupling between flame dynamics induced by buoyancy and fine fuel particles response to convection, where the ignition process is critically dependent on unsteady (pulsating) flame convection that make the physical contact with pre-heated fuel surfaces. The role of turbulence in fire dynamics (often based on large eddy simulations, LES) is a critical component (Sun et al., 2009) that is now embedded in advanced wildfire physical models that will be discussed shortly. Fuel-wind interactions are also important for various aspects of wildfire; Finney et al. (2015) have shown that convective cooling of fine-sized fuel particles by wind may offset heating by thermal radiation (see also Albini, 1986) until surfaces are convectively heated up by direct contact with flames and hot gasses. Additionally, in the following we will discuss the importance of fire spotting via airborne firebrands and embers facilitated by wind convection as an important process that contributes to long distance wildfire spread.

It is noteworthy that in parallel to progress in wildfire modeling in the US, Canada, Australia and Europe reviewed here, the Russian literature is replete with advanced mechanistic and empirical models that were developed independently, for example, the early work of Molchanov (1957), Grishin et al. (1981), Grishin (1997 compilation of wildfire research) and others reviewed by Goldammer and Furyaev (1996) in their monograph on *Fire in Ecosystems of Boreal Eurasia*.

3. Fire line intensity and flame shape

Byram (1959) introduced the concept of fire line intensity I_B (kW/m) as an integrative measure of wildfire characteristics that combines heat of combustion with fuel consumption and rate of fire spread. The fire line intensity represents the rate of energy (heat) release per unit length of fire front per unit time. Typically, Byram's fire line intensity is estimated from measured fuel consumption rate, heat of combustion and the rate of fire spread:

$$I_B = H_f \bullet w{\cdot}R \tag{6}$$

Where H_f is the fuel low heat of combustion (kJ kg^{-1}) (low indicates that the latent heat of vaporization of water in the reaction products is not recovered), w is the amount of fuel consumed during passage of the flaming front (kg m^{-2}), and R is the linear rate of fire spread (m s^{-1}). The images in Fig. 4 depict key geometrical ingredients that characterize fire line shape and are used to link field observations with Byram's fire line intensity metric. Alexander (1982) provides typical values for H_f that tend to vary within 10% among different natural fuel types with a representative value of 18,700 kJ kg^{-1}; w ≈ 1–5 kg m^{-2} where most of the variations in I_B are associated with fire's rate of spread R that may vary by two orders of magnitude (with wind, topography and fuel moisture content). The results depicted in Fig. 5 (Alexander and Cruz, 2019) show that fire line intensity can vary across 4 orders of magnitude from 10 to 100,000 kW/m (most intense crown fires fall within 10,000–30,000 kW m^{-1}). Alexander (1982) and the Canadian forest fire research community distinguish Byrams's frontal line intensity I_B (kW m^{-1}) from the rate of heat release per unit area in the active flame zone: $I_R = I_B/D$ (kW m^{-2}), with D the active flame depth (m, see Fig. 4a). Indirect estimates of I_B have been based on observations of flame length or crown scorch height. Byram (1959) established empirical relations

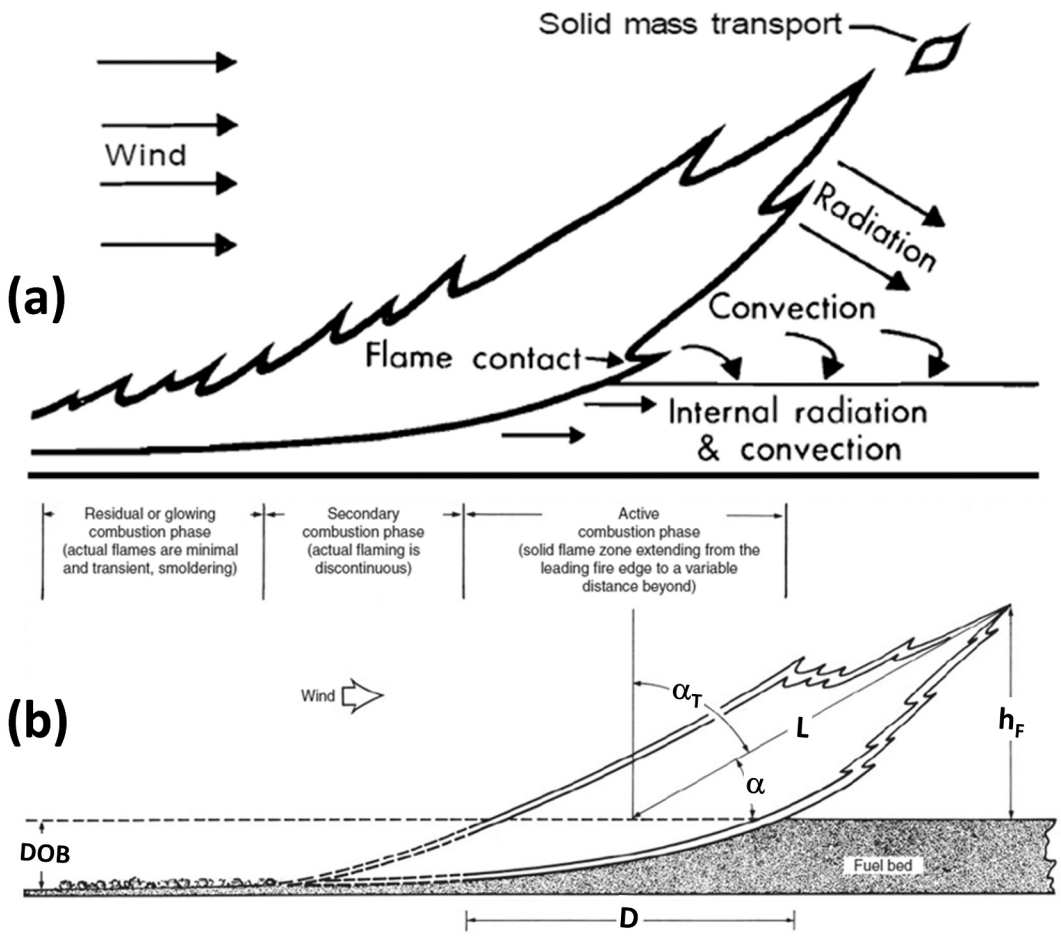


Fig. 3. A sketch of (a) Rothermel's surface fire spread model with key processes ahead of the fire frony (from Rothermel, 1972); (b) wind-driven fire front illustrating heat release during and following passage of the front with flame length (L), flame height (h_F), flame angle (α) and tilt angle (α_T), front width/depth (D) and depth of burn (DOB) (from Alexander, 1982).

between I_B (kW m^{-1}) and flame length L (m) given by (Alexander, 1982):

$$L = 0.0775 \cdot I_B^{0.46} \tag{7}$$

This empirical equation has been inverted and used to estimate fire line intensity from observations of flame length according to:

$$I_B = 259.8 \cdot L^{2.174} \tag{8}$$

Alexander and Cruz (2019) have compiled a list of different relationships between fire line intensity and flame length reported in the literature based on different fire and fuel types and various experimental settings conditions. More recently, Johnston et al. (2017) have made direct infrared measurements using the fire radiative power framework to remotely deduce I_B of spreading wildfires with promising results.

The passage of a fire line of different radiative intensities is an important event for wildfire spreading and for its impact on the landscape. We can view this via fire radiative energy (FRE) density that heats unburned fuel or soil surfaces, or via the induced air and surface temperatures such as depicted in Fig. 6. A typical trajectory of temperature evolution during passage of fire line in a pine forest is depicted in Fig. 6, where temperatures exceeding 100^0–200^0 C lasted for one minute only during this burn experiment. Radiative energy release measured by fire line intensity involves both fuel consumption and rate of fire spread, hence, similar intensities can result from different combinations of these two variables. Considering the release of energy during a wildfire, Alexander (1982) concluded that I_B (or I_R) alone is insufficient for assessing fire impact because after passage of a fire front considerable smoldering and burning may take place (see secondary combustion on Fig. 3b). Some of the most significant alterations of soil properties are associated with localized burnout after the relatively rapid passage of a fire front (Ulery et al., 1996; Certini, 2005).

The burning of large fuel elements during the so-called burn-out time t_B lasts longer than the fire front residence time (estimated as $t_R = D/R$, with D the fire line depth, see Fig. 4a). Estimates of burnout times t_B can be derived from knowledge of fuel element sizes (Fig. 7) or in practice, from net heat of combustion H_f (kJ kg^{-1}); total fuel consumed w_T (kg m^{-2}) divided by combustion rate C (kW m^{-2}):

$$t_B = H_f \cdot w_T / C \tag{9}$$

The combustion rate (heat release per area in unit time) is affected primarily by fuel size (Fig. 7). For certain fire regimes, C has been estimated from field studies via: $C = H_f w/t_R$ (Alexander, 1982), however, different estimates of C must be considered for estimation of burnout times in eq. 9 (note that w in Eq. 5 can be different than w_T for burnout time considerations). More nuanced solutions for the ignition and burning times of large fuel elements (tree trunks) have been studied by Albini and Reinhardt (1995) summarized in the "BURNUP" algorithm discussed below. Burrows (2001) reported measurements of residence and burnout times for different vegetation fuel sizes and types from Australia (Fig. 7). The different characteristic times (flame residence time vs. burnout time) and derived metrics are linked with characteristics of fuel heating and flame reaction and affect the combustion efficiency of wildfires, which has been shown in recent studies by Harmon et al. (2022) to be surprisingly low in stands (0.1 to 3.2%) and landscape

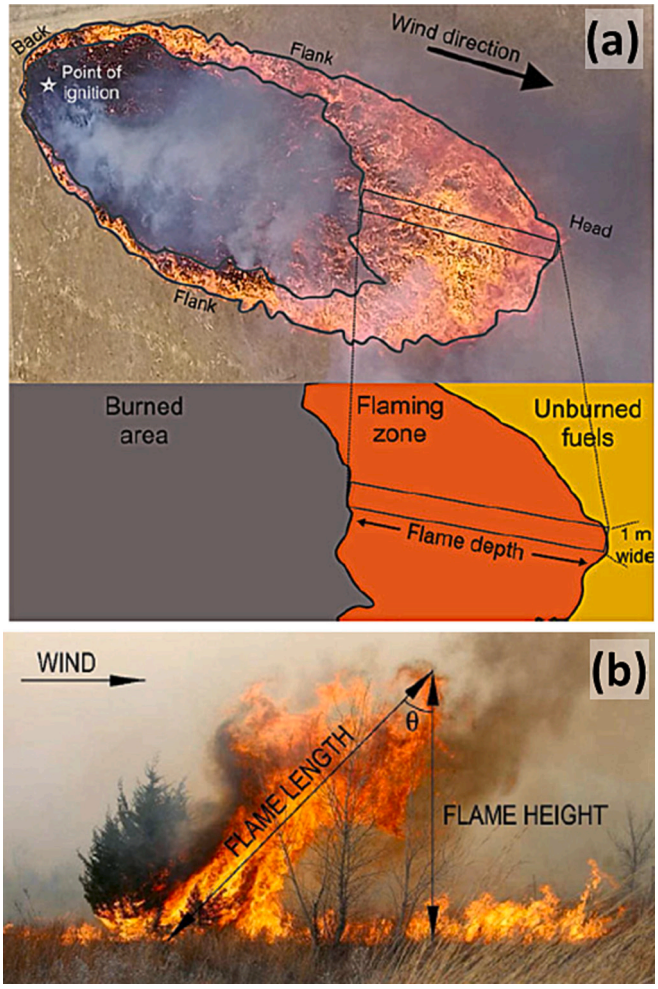


Fig. 4. Annotated images of (a) fire front area for calculation of Byram's fire line intensity (Alexander and Cruz, 2019); (b) cross section of fire front over flat terrain and definition of wildfire flame geometry (Egorova et al., 2022).

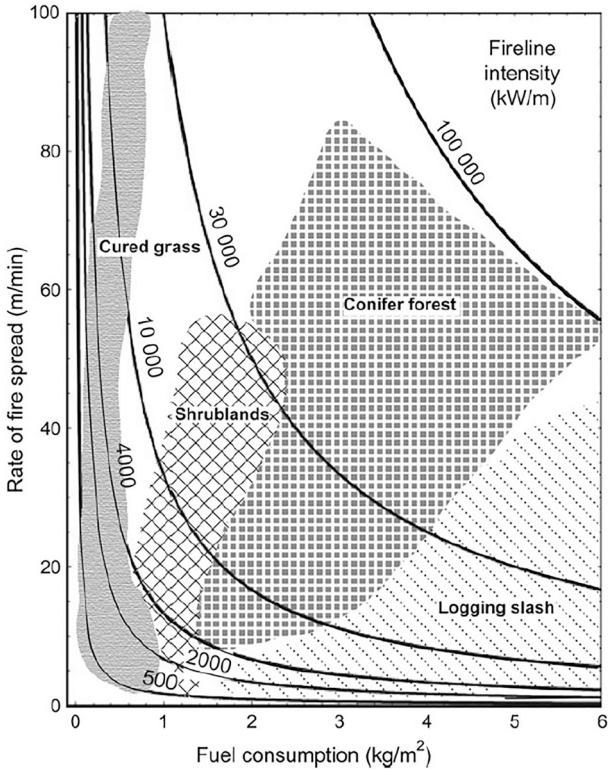


Fig. 5. Wildfire characteristics diagram depicting five fire line intensities as functions of rate of fire spread and fuel consumption. The calculations assume "net" fuel low heat of combustion of 18,000 kJ/kg and four fuel complexes: grass, shrubland, conifer forest and slash (Alexander and Cruz, 2019).

scales (0.6–1.8%). These low values are explained by the dominance of large trees with low combustion rates (Stenzel et al., 2019). In contrast with fire interactions with large fuel particles (branches and tree trunks), field data show that most of the fine fuel (90% of the leaves and needles) is consumed within the flaming zone, a distinction that greatly simplifies ignition and combustion modeling (Morvan and Dupuy, 2004). Even if combustion efficiencies are higher in wildfires, these recent reports shed light on how fuel structure affects energy release, the time scales involved, and the combustion efficiency that jointly determine how wildfires impact landscapes beyond Byram's fire line intensity metric.

4. Fire spotting via airborne firebrands and embers

An important mechanism for wildfire spread, particularly at the WUI is via showers of airborne firebrands that can travel considerable distances from the fire line and start new fire spots (Manzello et al., 2020). Fire spotting involves firebrands lofting into rising flame and combustion gases and being laterally transported via the ambient wind field until they fall under gravity and potentially initiate new spot fires (Albini et al., 2012). Efforts to quantify fire spotting and firebrand travel distances date back to early studies by Tarifa et al. (1965) and Albini (1979). Recently, Manzello et al. (2020) reviewed key processes involved in firebrand production and their subsequent transport. Firebrands are often generated during the thermal disintegration of pieces of fuel that may then be separated by airflow drag forces surrounding the burning fuel and lofted by rising plumes of hot air. The characteristics of firebrand generation depend on the type of fuel and its morphology and on fire intensity and the resulting buoyant plume. Models for firebrand generation and their flux density (firebrands/area) remain at its infancy. Recent experimental studies by Thomas et al. (2017) have established empirical estimates of firebrand flux density as a function of fire characteristics. Reviews by Manzello et al. (2020) and Thomas et al. (2021a) provide additional insights into the complex firebrand challenge as summarized in Fig. 8 that summarizes our present understanding of firebrand generation at the fire line and subsequent lofting by fire plume (Fig. 8a); experimental results of size and mass of firebrands (Fig. 8b), and firebrand deposition density as a function of distance from the fire line (Fig. 8c). Theoretical and observational advances are particularly important for assessing fire spread risks at the WUI in which direct fire and firebrand exposure are the leading causes of the ignition of structures.

For fire spreading beyond fire line advance rates, firebrands present an important mechanism and a challenge to the prediction of wildfire behavior and patterns. Firebrand long distance transport complicates fire control and suppression operations with the potential formation of new fire spots a few kilometers ahead of the fire line. Transport trajectories and distances of firebrands have been studied extensively using a range of mechanistic models such as developed by Albini and coworkers, statistical-mechanistic models of Egorova et al. (2022) and advanced large eddy simulation (LES) model of Houssami et al. (2016) and others. For example, Fig. 9 depicts model predictions for the maximum firebrand and fire spotting distances as a function of wind speed for different fire types (Albini et al., 2012). Studies have shown that for embers of equal initial mass, disk shaped embers propagate farthest and have the highest remaining mass fraction upon landing; spherical embers are carried the shortest distance, and cylinders have the smallest mass

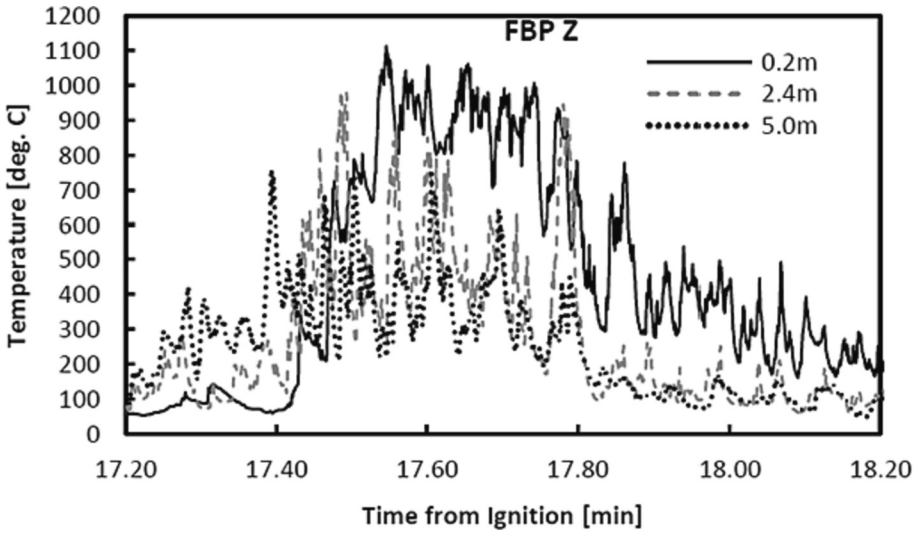


Fig. 6. Temperature evolution during passage of fire line through a burned parcel in Pinelands National Reserve, New Jersey, dominated by pitch pine and intermittent oaks. Measurement using K-type thermocouples installed at several heights above the ground (from Thomas et al., 2017).

fraction upon landing (Anthenien et al., 2006).

Despite inherent limitations and simplifying assumptions, the relatively simple and physically-based models described above have been instrumental in transforming wildfire science from an empirical discipline to a modern scientific field with considerable predictive capabilities, systematic metrics and a long legacy of close coupling between observations and modeling. Limitations of small-scale (wind tunnel) experiments and limits to scalability of certain components such the proportionality of radiation to flame size (Egorova et al., 2022) are recognized and considered in dimensional analyses that remain essential to improving predictions. In the following we will review the leading wildfire simulation models used in research and practice and highlight principles, strengths, and limitations.

5. Classification of wildfire models

In the following, we seek to provide an overview of the key wildfire models used in research and in practice and summarize their characteristics. The classification of wildfire models in the literature varies considerably, for example, Weber (1991) designated "empirical" all models that do not distinguish among forms of energy exchange (radiation and convection). Mell et al. (2018) considered only models that employ computational fluid dynamics (CFD) as "physical" models (see exchange with Cruz et al., 2017; Cruz and Sullivan, 2018). Here, we consider three broad types of models for predicting wildfire characteristics, specifically the rate and shape of fire line spreading: (i) statistical-empirical, (ii) semi-physical and (iii) physical-mechanistic. We focus on delineating operational models that represent the entire process of wildfire behavior and fire line propagation including software packages derived from these models.

(i) ***Statistical-empirical models***: Models that are based on statistical correlations among fire-related quantities (McArthur, 1966) and newer models resulting from application of machine learning as *statistical-empirical*. Perry (1998) defines this class of models as: "*Empirical models are in essence statistical descriptions of wildland fires and make no attempt to include any of the physical mechanisms which drive the fire process*".

(ii) ***Semi-physical models***: The next level of mechanistic representation of fire behavior invokes energy conservation. These models consider salient features of fire line physics and attempt to predict fire line shape in two-dimensional space. Semi-physical models are often based on information from small-scale experiments and employ dimensional analysis for similarity across scales (with limitations of small-scale experiments and scale-dependent radiation).

(iii) ***Physical-mechanistic models***: are based on mathematical analysis of the fundamental physical and chemical processes of fire spread and are independent of analyses of actual fire events. We follow Weber (1991) and Perry (1998) model classifications based on the three categories listed here, however, in Table 4 we expand Weber (1991) and Perry's (1998) classification by adding a summary of physical simulation models presented in Morvan (2011) and process coupling and scales presented in Bakhshaii and Johnson (2019). In this review, we discuss additional operational software packages derived from the physical models in Table 3. Some have proposed to distinguish between the core physical model and the software packages used for fire simulations (Sullivan, 2009a, 2009c), we will maintain this distinction in the review of physical and semi-physical models and add new models that address wildfire behavior and spread from basic principles.

(iv) ***Other models***: wildfire spread models derived from analogy to similar phenomena from cellular automata to percolation theory and diffusion reaction models (Sullivan, 2009c) are also addressed here. These models harness similarity and principles that capture some of the patterns and spreading of wildfires but (typically) do not consider details of wildfire physics. Unlike Perry (1998), we do not classify these as "empirical" due to their reliance on physical principles and use of physical principles to derive fire distribution rules such as in the Quic-Fire model (Linn et al., 2020).

6. Statistical-empirical wildfire models

Statistical-empirical wildfire models were developed in Australia (McArthur, 1966, 1967; Noble et al., 1980). These early models were based on an extensive database of 5000 uncontrolled fires, and a few (500) intensively studied prescribed burns summarized in nomograms for fire hazards or simple empirical equations for fire danger and spread rate derived from this fire danger index. Typically, a fire danger index, F, integrates air temperature (T_a,^{0}C), relative humidity (RH,%) wind speed at 10 m height (U_{10}, m/s) and the degree of fuel curing (C_d,%) in an (empirical) expression such as (for grassland):

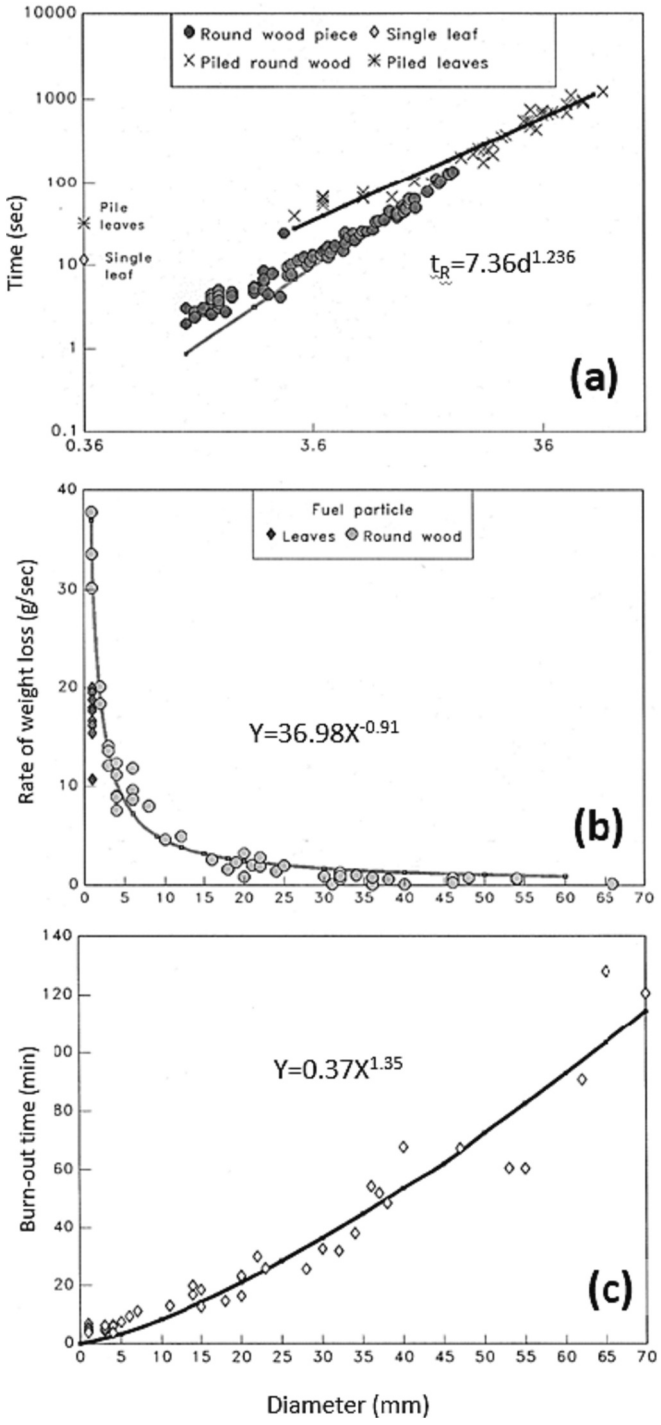


Fig. 7. Fuel diameter affecting flame residence time, weight loss and burnout times in eucalypt: (a) flame residence time for single fuel pieces (d = diameter); (b) rate of weight loss during flaming combustion of 1000 g of fuel burning on a load cell; (c) time taken for complete combustion (burn-out) of dry round wood particles (from Burrows, 2001).

$$F = 2 \bullet exp\left(-23.6 + 5.01 \cdot ln(C_d) + 0.0281 \cdot T_a - 0.226 \cdot RH^{\frac{1}{2}} + 0.633 \cdot U_{10}^{\frac{1}{2}} \right) \quad (10)$$

with the rate of fire spread R (km hr^{-1}) given by $R = 0.13F$. Refinements to account for fuel type (e.g., grass), fuel moisture content (%) and drought factors are given in Noble et al. (1980). Note, the degree of curing refers to the percentage of senesced or dead grass in the fuel bed.

More recent applications of this class of statistical-empirical models (e.g., Cheney et al., 1998) have introduced refinements into the relationships that build upon information and insights gained over the years. Perry (1998) noted that the application of these empirical models to ecosystems other than the ones used in their derivation was not very successful. In an extensive review of empirical models, Sullivan (2009b) lists statistical-empirical models from around the world and their applications.

Beyond the classical statistical wildfire models (based on correlations and regressions with measurable variables), there is a new class of empirical (data-driven) wildfire models based on application of machine learning (ML) and artificial intelligence (AI) methods. Jain et al. (2020) provided an extensive review of various ML algorithms used for wildfire modeling. Different methods have been applied to address various aspects of the wildfire problem such as: (i) fuels characterization, fire detection, and mapping; (ii) fire links to climate change; (iii) fire occurrence, susceptibility, and risk; (iv) fire behavior prediction; (v) fire effects; and (vi) fire management. Among these, we consider ML applications to modeling fire behavior as most relevant for the purposes of this review. A recent ML-based wildfire spread model in grasslands was developed by Khanmohammadi et al. (2022) using support vector regression (SVR), Gaussian process regression (GPR), regression trees and neural networks (NN) algorithms (Fig. 10). Interestingly, the ML model uses similar input variables as the original model of McArthur (1966) to predict rates of fire spread. These recent ML applications for wildfire behavior remain largely exploratory focusing primarily on the performance of various ML algorithms (Khanmohammadi et al., 2022). Coffield et al. (2019) demonstrated the ability of decision trees to predict final wildfire size from ignition points and meteorological data.

7. Semi-physical wildfire models

A class of fire spread models strike a compromise between the complexity and rigor of physical models (that require description of turbulent flows and combustion reaction kinetics) and models where fire behavior and heat fluxes are derived experimentally (Rothermel, 1972). These semi-physical models are often used to represent wildfire behavior over landscapes (two-dimensions) by converting 1-D forward spreading wildfire models into two dimensions assuming prescribed shapes (ellipses) and rules for how the fire perimeter evolves. Semi-physical models can also be distinguished based on whether they use a regular grid system (also termed *point techniques*) or models that represent the fire as a continuous object on a plane (Richards, 1995). For example, fully physical fire spread models (e.g., *WFDS, FIRETEC, FIRE-STAR*) and cellular automata fire models (e.g., *QUIC-Fire*; Linn et al., 2020) often use a regular grid over the simulation domain to spatially discretize the partial differential equations (or create a cellular domain). In contrast, continuous line fire spread models (e.g., *FARSITE*, Finney, 1998) use prescribed shapes such as ellipses or other evolving shape models (Richards, 1995).

Several semi-physical fire behavior models formulate Rothermel's (1972) fire line, crown fire behavior (Van Wagner, 1977), and spotting processes (Albini, 1979) as point sources on the fire line perimeter (prescribing rates and distances of spread from a given location) that lead to vector fire perimeters produced by Huygens' principle, a concept borrowed from optical wave propagation to represent localized sources for fire spread (Fig. 11 a and b). In essence, the Huygens' principle with point vectors, vertices or markers that are connected to form a fire front give rise to a Lagrangian framework with rapidly mounting computational costs as the fire expands (Bova et al., 2016). An alternative approach is based on the "Eulerian level set" method for tracking expanding and merging fire fronts including natural emergence of unburned fuel islands due to multiple ignition points such as emerging from fire spotting (Lautenberger, 2013). The Eulerian level set method has been implemented in *WRF-Fire* and *WFDS*, and recently tested by Bova et al. (2016) showing similar results to the Lagragian fire line

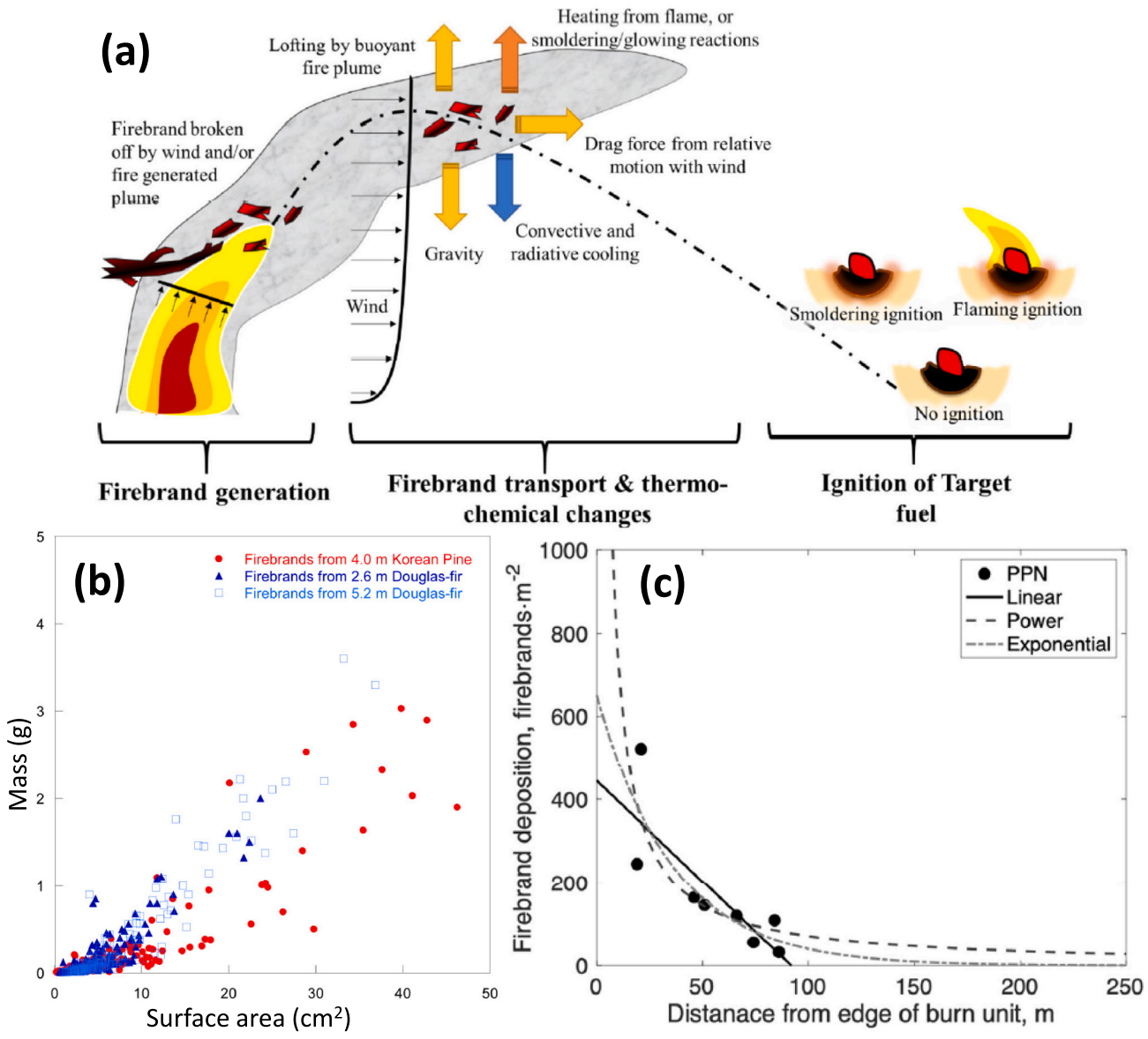


Fig. 8. (a) Sketch of firebrand generation, transport and deposition (from Manzello et al., 2020); (b) Experimental data of surface area and mass of firebrands (from Manzello et al., 2020); (c) Firebrand density vs distance from burn surface (from Thomas et al., 2021b).

propagation (e.g., *FARSITE*) as depicted in Fig. 11 c suggesting equivalence of the methods for most wildfire scenarios of practical interest.

In the following we provide a brief overview of three widely used semi-physical models – *BEHAVE* (Andrews, 2014) and related *FARSITE* (Finney, 1998), the Canadian *Prometheus* (Tymstra et al., 2010), and the Australian *SiroFire* (Coleman and Sullivan, 1996).

7.1. *BEHAVE and BEHAVE_Plus (Andrews, 2014)*

The National Fire Danger Rating System (NFDRS) and the *BEHAVE* fire prediction system (Andrews, 2014) are based on the Rothermel (1972) fire front spreading model in a landscape specified by fuel, weather and topography. The *BHEHAVE* model has evolved to include fire spread and a fuel model that treat the spread of fire as a series of ignitions where the heat transfer from each successive strip of fuel raises the next strip to ignition temperature, thus propagating the fire. *BEHAVE* has been used extensively for training purposes and for prescribed burning planning. In 2002 the *BEHAVE* model was updated to *BehavePlus* by integrating the state of knowledge to more than 40 different fire modules (Andrews, 2014), well beyond Rothermel's original fire spread model. The fire spread model is now in the core of the Surface module (Fig. 12) that considers steady advance of a fire front over a uniform fuel bed independent of ignition details. The *Surface* module implements Byram's (1959) fire line intensity and flame length as well as Albini's (1976) formulation for Rothermel's model with explicit parameterization of fuel characteristics and moisture. The *BehavePlus* model considers wind conditions, critical fire intensities for transitioning to crown fires and their subsequent spread rates, and onset of spotting distances for different fire regimes (Andrews, 2014). Certain processes and fire-related phenomena are lacking in the current *BehavePlus* such as smoke production, and scorch height for crown fire transitions, importantly, post-frontal fuel consumption such as large fuel burnout are lacking (they are included in *FARSITE* discussed next). Some of these omissions are critical for assessing wildfire impacts on soil properties and on post-fire hydrology discussed later.

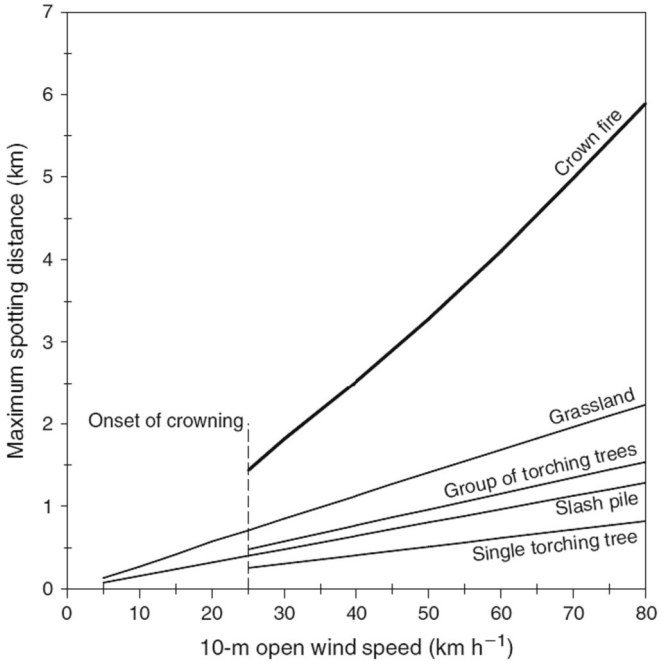


Fig. 9. Model predictions of maximum fire spotting distances over level terrain as a function of wind speed for specified burning conditions (from Albini et al., 2012).

Table 4
Range of temporal and spatial scales for primary physical and chemical phenomena involved in wildfire behavior (from Sullivan, 2009a).

Type	Time Scale (s)	Vertical Scale (m)	Horizontal Scale (m)
Combustion reactions	10^{-20} - 10^{2}	10^{-4}–10^{-2}	10^{-4}–10^{-2}
Fuel particles	–	10^{-3}–10^{-2}	10^{-3}–10^{-2}
Fuel complex	–	1–20	1–10^{2}
Flames	10^{-2}–30	0.1–10	0.1–2
Radiation	10^{-9} - 10	0.1–10	0.1–50
Conduction	10^{-2}–10	10^{-2}–10	10^{-2}–10
Convection	1–10^{2}	0.1–10^{2}	0.1–10
Turbulence	10^{-1}–10^{-3}	1–10^{3}	1–10^{3}
Spotting	1–10^{3}	0-3 × 10^{3}	1–10^{5}
Plume	1–10^{5}	1–10^{5}	1–10^{2}
Combustion reactions	10^{-20} - 10^{2}	10^{-4}–10^{-2}	10^{-4}–10^{-2}
Fuel particles	–	10^{-3}–10^{-2}	10^{-3}–10^{-2}
Fuel complex	–	1–20	1–10^{2}
Flames	10^{-2}–30	0.1–10	0.1–2
Radiation	10^{-9} - 10	0.1–10	0.1–50
Conduction	10^{-2}–10	10^{-2}–10	10^{-2}–10
Convection	1–10^{2}	0.1–10^{2}	0.1–10
Turbulence	10^{-1}–10^{-3}	1–10^{3}	1–10^{3}
Spotting	1–10^{3}	0-3 × 10^{3}	1–10^{5}
Plume	1–10^{5}	1–10^{5}	1–10^{2}

7.2. *FARSITE (Finney, 1998; revised 2004)*

The Fire Area Simulator, *FARSITE* (Finney, 1998; Finney and Andrews, 1999) incorporates key modeling elements from *BEHAVE* (primarily the Rothermel fire spread module) and from the Canadian Forest Fire Behavior Prediction System (a precursor to the Prometheus, CWFGM Steering Committee, 2004). *FARSITE* is strongly coupled with spatial geographic information systems (GIS) and has been used by state and federal agencies for prediction and management of prescribed burns, training courses, evaluating fuel treatments (Finney, 2001; Finney, 2006) and reconstructing past fires (Duncan and Schmalzer, 2004; Price and Germino, 2022). *FARSITE* requires a minimum of five GIS raster layers to generate simulations including: elevation, aspect, slope, fuel model, and canopy cover. Fire propagation follows a vector approach (Fig. 12 a and b) for the position of the fire front at specified times based on the Huygens principle. While rasters are used to represent the underlying landscape and to record fire characteristics during the simulation, the fire perimeters are processed and stored as continuous vectors. The *FARSITE* model has been used by the US Forest Service and the National Park Service for long-range projections of wildland fires. The latest revision of *FARSITE* incorporates modules for surface fire, crown fire, spotting, fire acceleration, and fuel moisture (Finney, 1998). More recent additions to FARSITE include the use of spatially resolved and rasterized wind fields derived from an external program *WindNinja* (Wagenbrenner et al., 2016).

7.3. *Prometheus (CWFGM Steering Committee, 2004; Tymstra et al., 2010)*

The Canadian Wildland Fire Growth Model, Prometheus (CWFGM Project Steering Committee, 2009) is based on the Canadian Fire Behavior Prediction (FBP) System and uses the wavelet propagation algorithms of Richards (1995) to simulate the spread of wildland fire across landscapes (similar to *FARSITE*). Prometheus is Windows-based and uses maps and geographic data exported from GIS platforms for real-time wildfire operational management tool. The inputs for weather and fuel are according to the Canadian Fire Behavior Prediction (FBP) system. Originally, only three fuel types were used that do not cover all possible Canadian fuel types. Tymstra et al. (2010) report that newer versions of Prometheus use 16 standard fuel types of the Canadian FBP with fairly complex look up tables for composition and conditions and grid modifiers. Weather streams can be entered from observations, forecasts or generated based on average conditions for the time of year and latitude.

7.4. *Australian National Bushfire Model (Coleman and Sullivan, 1996) and SiroFire*

Unlike *FARSITE* and *Prometheus* that were developed for forest fires in North America, the Australian Bush Fire Model (later evolved to *SiroFire*) is rooted in the early empirical grassland fires and the so-called meters for modeling fire spread developed by McArthur (1966). The *SiroFire* simulator (Coleman and Sullivan, 1996), calculates fire growth using different fire behavior models. Like *FARSITE* and Prometheus, the fire line is propagated using an assumed ellipse based on local fuel, slope, weather and specifics of the fire behavior model. *SiroFire* allows for temporal variations in temperature, relative humidity, wind speed and direction (that affect fire advance and fuel moisture content). *SiroFire* considers two fuel types with spatial input of fuel load (for forest in the range of 5 to 40 ton/ha). For grass, users can spatially vary the curing percentage and the grass type (no grass, eaten out, grazed, ungrazed). Like Prometheus, *SiroFire* does not propagate crown or spot fires. There are no plans to continue development of *SiroFire* with version 2.5a last updated in 1998.

8. Physical wildfire models

The conditions for fuel ignition and combustion are largely predictable. However, the predictability of wildfires that consume complex natural fuels under shifting weather and often unknown fuel conditions is far less certain. Every wildfire event is unique and non-repeatable (much like a specific landslide or an earthquake event). Adding to the spatial heterogeneity and local weather uncertainties, the physical and chemical processes involved in wildfire modeling span across many scales (see Table 4 from Sullivan, 2009a) with various key processes interacting nonlinearly. Considering these inherent non-linear phenomena (turbulence, combustion, radiation) involved in wildfire

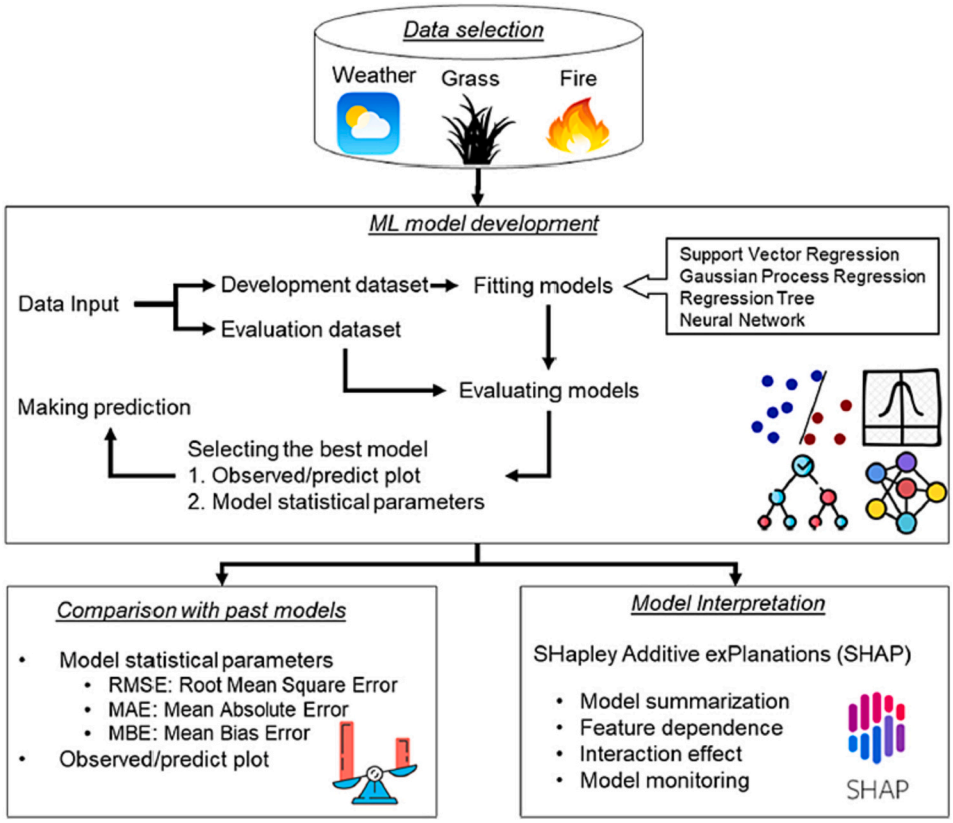


Fig. 10. A machine learning approach for predicting rate of fire spread trained on observations and empirical relations (from Khanmohammadi et al., 2022).

behavior, some argue that no general statements regarding model accuracy can be made at present state of knowledge (Morvan, 2011). Nevertheless, a century of systematic wildfire research and the unprecedented expansion of spatial and temporal observation capabilities offer new opportunities for quantitative and predictive representation of this complex natural phenomenon.

The key processes and components for wildfire physical modeling are well known (Perry, 1998) and are routinely implemented for prediction of wildfire behavior (Fig. 13). In the following we review key physical models used in research for hypothesis testing and for selected field applications (i.e., prescribed burning and small-scale well characterized fires).

Fully physical models have several characteristics in common (see review by Morvan, 2011 and discussion by Mell et al., 2018); (i) they explicitly compute the dynamic interactions between ignition, fuel consumption, and atmospheric flows at the fire line often using computational fluid dynamics (CFD) with large eddy simulation (LES) modules for expressing mixing and coupled atmospheric flows with fire line feedbacks at high spatial and temporal resolution (Linn et al., 2002; Mell et al., 2007); (ii) they solve the multiphase exchanges (conversion of solid fuel to gas products and charcoal with temperature) including combustion reaction rates, convective and radiative exchanges and spatial averaging (subgrid representation) required for computations and parameterization.

8.1. *FDS and WFDS Models (McGrattan et al., 2010; Mell et al., 2007)*

The Fire Dynamics Simulator (*FDS*) was originally developed by the National Institute of Technology and Standards (NIST, McGrattan et al., 2010) for fire safety simulations. It is a CFD model of fire driven fluid flow that solves the Navier-Stokes equation for low-speed flows with emphasis on smoke and heat transport from fires. *FDS* uses a three-dimensional rectilinear grid to solve mass, momentum, and energy conservation equations for combustion with LES. The idea behind using LES is to calculate the role of eddies that are responsible for mixing and transport of gases at reasonable accuracy from fluid dynamics (Navier-Stokes equation) thus capturing interactions (flow and combustion) at high spatial and temporal resolution. Using the same grid used for fluid flow, thermal radiation is calculated using the finite volume method. Lagrangian particles (typically soot produced from chemical reactions represented in *FDS*) are used to simulate smoke movement (*Smokeview* is a companion program used for visualizing smoke and fire). *FDS* was developed primarily for questions related to fire protection engineering and to study fire dynamics and combustion. The user provides all inputs via a single text file that contains information about the numerical grid, ambient environment, building geometry, material properties, combustion kinetics, and desired output quantities (gas temperature, smoke concentration, heat release, pressure, etc.).

The Wildland Fire Dynamics Simulator (*WFDS,* Mell et al., 2007) is an extension of the *FDS* to large spatial scales with the objective of predicting wildfires and smoke transport at the wildland-urban interface (WUI) where natural fuel (forest, grassland) interacts with combustible man-made structures. WFDS has been used as a research tool with various computational grids capable of resolving volumes as low as 3.5 m^3 in a simulation domain of the order of 2.25 km^2 in area and 200 m high (Mell et al., 2007). The multi-grid discretization is used to resolve the LES equations and is equipped with a level set algorithm (Eulerian framework) for spatial representation of propagation of the fire front. *WFDS* uses temperature-dependent mass loss rate expression of Morvan and Dupuy (2004) to model solid fuel degradation assuming pyrolysis begins at 400 K (see Fig. 14 below). Combustion occurs solely due to fuel gas and oxygen mixing in stoichiometric proportion (independent of temperature – unlike the *FIRETEC* model). While details of the chemical kinetics are not resolved in the *WFDS,* the model considers the mass ratio of gas fuel to oxygen to compute mass loss rates (assuming chemical reactions are faster than mixing – hence turbulent mixing limit or

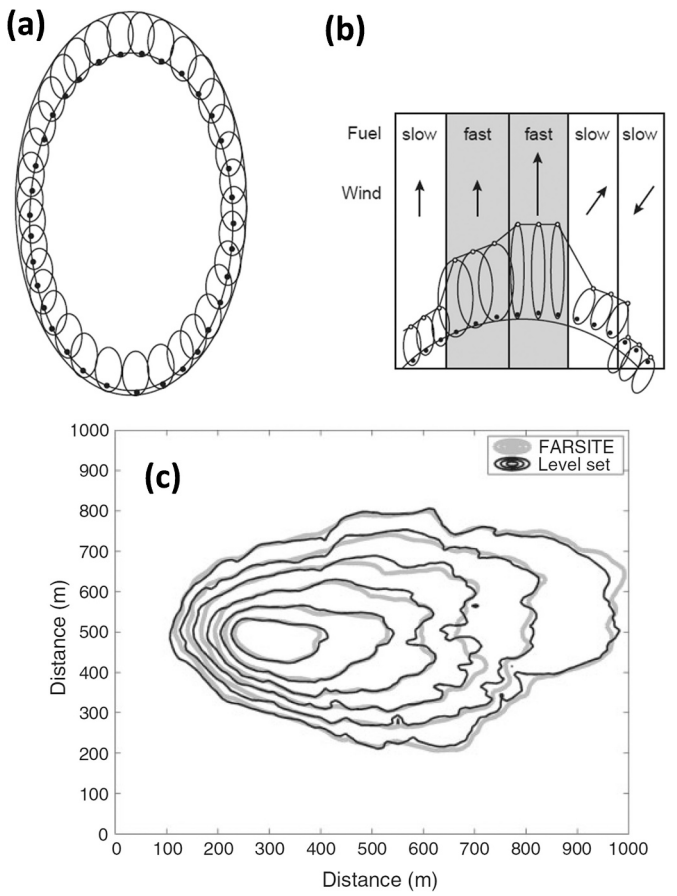


Fig. 11. **(a)** Illustration of Huygens' principle applied to spread of fire using elliptical wavelets for uniform conditions with wavelets (ellipses) of constant shape and size that maintain overall elliptical fire line shape over time (Finney, 1998); **(b)** Nonuniform conditions that schematically show how local ellipses size and orientations vary with local fuel type and wind vector (Finney, 2004); **(c)** Simulated fire line contours for grass fire by *FARSITE* (grey) in comparison with Eulerian level set predictions (black) for 18 km/h wind left to right (Bova et al., 2016).

control reaction rate). The *WFDS* (and its source *FDS*) is an LES model that invokes low Mach number approximation with an efficient pressure solver that reduces computation time. Presently, *WFDS* is not maintained as an independent model, interested users are referred to the well-documented and continually maintained *FDS* package.

8.2. *FIRETEC Model (Linn, 1997)*

FIRETEC, developed at the Los Alamos National Laboratory, is a three-dimensional wildfire model (Linn, 1997) that conserves mass, momentum and energy employing fully compressible gas transport for the combustion, heat transfer and fluid flows using high gradient flows (*HIGRAD* module; Linn and Cunningham, 2005). Numerically, *FIRETEC* uses the finite volume method to solve the evolving set of equations without necessarily resolving certain small-scale processes due to original design as a wildfire model with a relatively coarse grid (1 to 4 m, Linn et al., 2002). Consequently, certain computational compromises were made, for example, pyrolysis and combustion were treated as one step at the same location without resolving transport of different pyrolysis gases. In a later study, Colman and Linn (2007) have introduced refinements to the pyrolysis and combustion steps in *FIRETEC*. *FIRETEC* focuses on representing the average behavior of quantities that enter the conservation equations (within the smallest "resolved volume", hence details of fuel structure and flame fluctuations are not explicitly

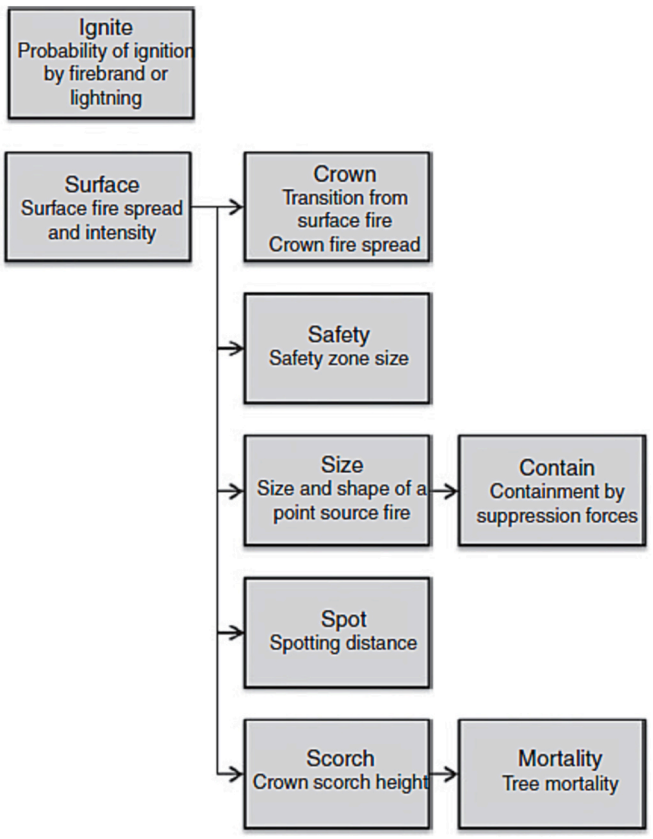


Fig. 12. Core modules in ***BehavePlus*** that can be used independently or linked where output from one module serves as input to another (From Andrews, 2014).

represented while quantities such as average concentrations and temperatures of the constituents (solid and gases) and other variables are expressed as means and fluctuations for ensemble averaging. The complex wildfire chemical reactions are reduced in *FIRETEC* to a single step combustion scheme:

$$Solid_{Fuel} + N_S O_2 \rightarrow (1 + N_S)\ Products + Heat \quad (11)$$

With N_S the stoichiometric coefficient (mass of O_2 for burning 1 kg solid fuel). Clark et al. (2010) have introduced a subgrid thermodynamic equilibrium refinement to address unresolved combustion processes in the "resolved volume". Once the mean temperature in a volume exceeds critical temperature (set at 500 K, Linn, 1997), combustion begins, and the evolution equations are used to track gas and solid species and energy production.

A primary strength of *FIRETEC* is the explicit representation of turbulence in three dimensions using multi-scale LES with spectra corresponding to three dominant scales of typical fuel considering a full tree, branches, and leaves/needles. The resulting interacting flow field enables realistic and fire-affected representation of pyrolysis, solid-gas, and gas-gas reactions, as well as heat transfer. Conservation of mass and superb transport representation (including temperature tracking) makes it possible to estimate probabilities of firebrands and subsequent travel distances (Koo et al., 2012). The radiative transfer from the heated soot and gases is treated via a modified diffusion method (originally developed for parameterizing climate models) in which emitted radiative energy is absorbed by solid objects based on their projected surface area per volume. The radiative scheme in *FIRETEC* is acknowledged to be inaccurate (Linn et al., 2002; Morvan, 2011) and various improvements have been made over time. Perhaps the most limiting aspect of *FIRETEC* is the choice of fully compressible solution of the Navier-Stokes equations instead of the low Mach approximation used by other fully

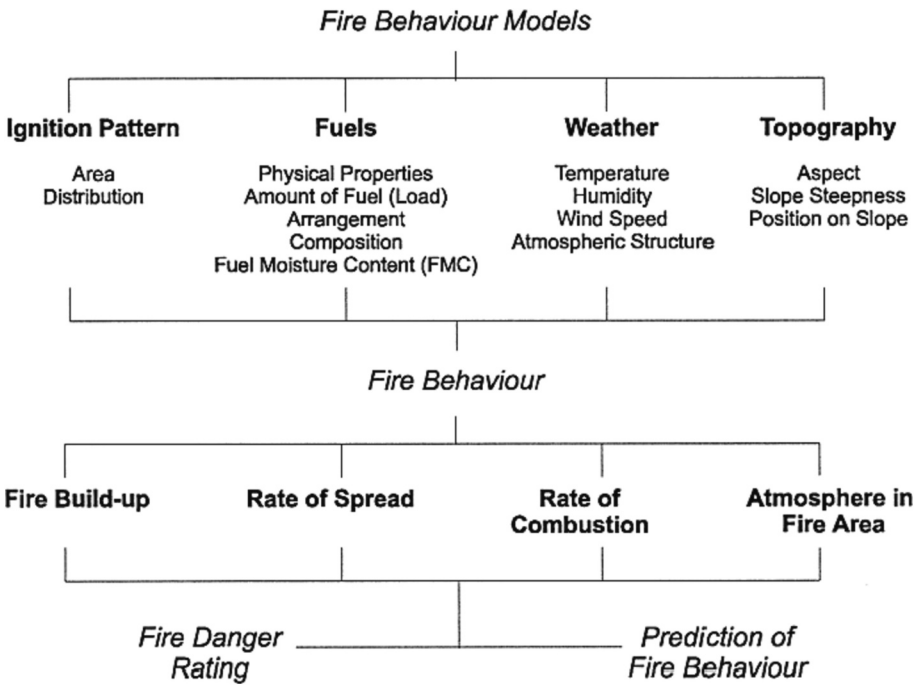


Fig. 13. A summary of key processes and components used for modeling wildfire behavior (Perry, 1998).

physical models (*WFDS, FIRESTAR*). The computational costs of the fully compressible solution for similar (subsonic flows) are time steps that scale with the Mach number (Morvan, 2011). Mell et al. (2005) found the *WFDS* to be far more efficient in terms of computational time compared with *FIRETEC* (largely due to the low-Mach-number formulation in *WFDS* and the very efficient Poisson solver used to calculate the pressure correction). For illustrative purposes we use an outdated benchmark reported by Colman and Linn (2007) that used 64 processors to run *FIRETEC* simulations considering 160 × 160 × 41 cells at a uniform horizontal grid resolution of 2 m (vertically nonuniform from 1.5 to 30 m) it took 1–2 min to compute every second of simulation time (more recent studies by Atchley et al., 2021 estimate 0.8 min computation time per second of simulation time).

8.3. *FIRESTAR Model (Morvan et al., 2006)*

FIRESTAR is a fully physical, two-dimensional (x-z) fire behavior model focusing on fire front dynamics and energy release (Morvan et al., 2006; Morvan et al., 2009). It considers fuel volume fraction, density, moisture content, and composition solved on one grid and fluid dynamics of the air and gaseous products of the solid phase tracked via a set of conservation equations for mass, momentum, and energy. Interactions between solid and gas grids are handled via coupling terms (aerodynamic drag, heat and mass transfer between the grids). Turbulence is handled differently than the models above, unlike the use of LES in *FIRETEC* and *WFDS, FIRESTARS* uses mass-weighted (transient) Reynold Averaged Navier-Stokes (RANS) representation of fluid flows. Combustion is governed by turbulent energy dissipation where eddy motion (responsible for gas mixing) and dissipation control reaction rates with provisions made for contributions of different eddy scales in the turbulent energy budget. Importantly, *FIRESTAR* considers solid fuel structure by separating thermally thick from thin fuel particles (diameter < 6 mm and surface to volume ratio $s_S > 600$ m^{-1}) for which internal thermal gradients can be neglected. This is important for the dynamics of thermal degradation and combustion (90% of the thin fuel is consumed during passage of the flame front) and affects the residence time and degree of combustion (Burrows, 2001; see Fig. 8 above). Experimental data and the model of Morvan and Dupuy (2004) depicted

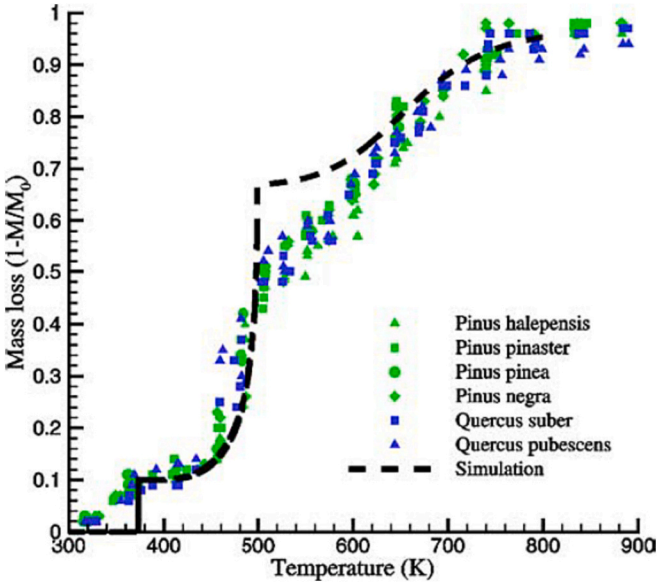


Fig. 14. Thermogravimetric analyses of various Mediterranean vegetation species and the pyrolysis model of Morvan and Dupuy (2004) used in FIRESTAR and WFDS models (1-M/M$_0$ represents relative mass loss).

in Fig. 14 show the critical temperatures of the thermal degradation of vegetation fuel, from evaporation of water at 373 K to pyrolysis at 400–500 K (formation of volatile gases) followed by combustion >500 K.

An important process that *FIRESTAR* resolves more rigorously than other models is the coupling of convective and radiative heat transfer between the flame and unburned fuel. Evidence suggests that radiation contributes at least 30% of the energy intercepted by unburned vegetation ahead of the fire front. *FIRESTAR* calculates the omnidirectional total irradiance from grid volumes including the contribution of flames (soot particles) and the important role of temperature fluctuations. Soot particle production, volume fraction and transport are calculated (for

simplicity soot particles are assumed to be spherical with diameter of 1 mm). The radiative transport model is solved using the *discrete ordinate method* (DOM) (i.e., discretizing irradiance in space and numerically integrating). *FIRESTAR* uses dynamic re-meshing with mesh size in the flame region is selected as a fraction of radiative transfer length (0.4–1.0 m) and larger mesh size used away from the fire line for computational efficiency (Morvan and Dupuy, 2004).

The rigor of fire-fuel interactions and highly resolved key processes in FIRESTAR make it an ideal model for illustrating some of the concepts regarding fire-atmosphere interactions or effects of fuel moisture content discussed above. For example, simulation results from grass fire reported in Morvan et al. (2009) highlight the effects of the fire line on wind profiles expressed as scaled horizontal velocity field before and after the fire front (points P1 and P2) in Fig. 15. The wind profiles are

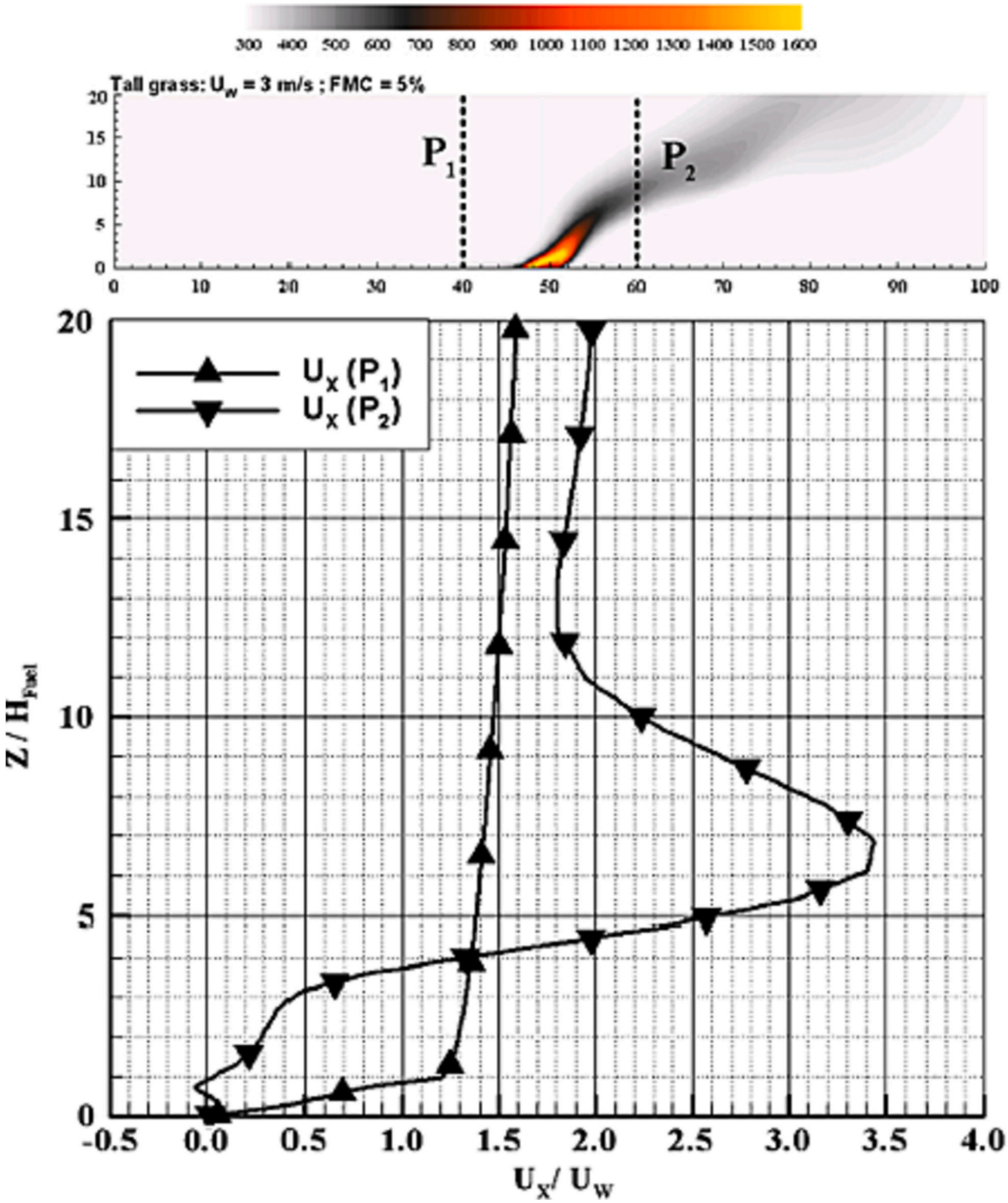


Fig. 15. The vertical distribution of longitudinal wind velocity scaled by wind speed at 2 m of U_w = 3 m/s (height is scaled by fuel height H_{fuel}). (top) simulation results of fire line shape and wind sampling locations (from Morvan et al., 2009).

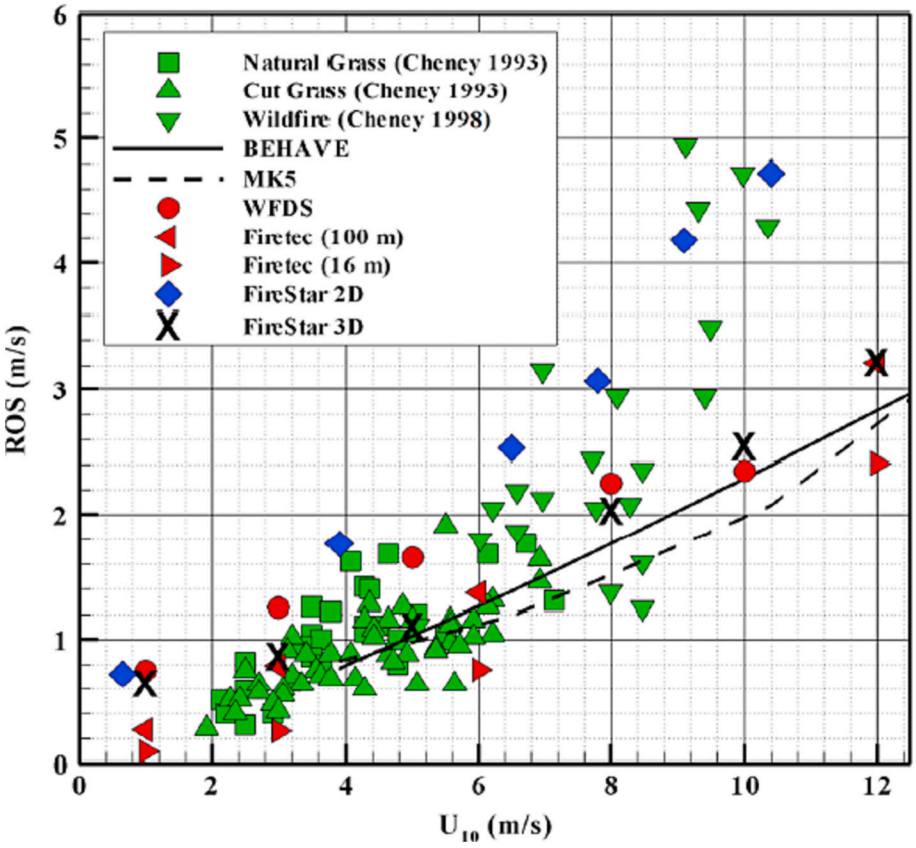


Fig. 16. Model intercomparison of rate of spread (ROS) of surface fire through grassland versus 10 m wind speed (U_{10}) (from Meradji et al., 2016).

scaled by 2 m free stream wind speed and by fuel height (0.7 m) in the z coordinate. The results show the effect of the fire line on air flow (for relatively low wind speed in a regime known as "plume dominated fire") with strong fresh air entrainment ahead of the fire front (wind profile at P_2 shows near surface air flowing at smaller than ambient wind speed). The same study compares rates of fire spread (ROS) in grass predicted by *FIRESTAR* and the Australian empirical model of Cheney et al. (1998) termed here *MK5* and *BEHAVE* semi-empirical model of Andrews (2018) for the same initial and boundary conditions. Morvan et al. (2009) have shown that all models exhibit similar results for changes in Byram fire line intensity with wind speed. The simulated ROS with wind speed varies among the models as depicted in Fig. 16 (data from wildfires and controlled grass fire experiments of Cheney et al., 1993). *FIRETEC* and *WFDS* predictions of ROS agreed with *BEHAVE* and *MK5*, whereas *FIRESTAR* (present 2D version) exhibited consistently high ROS. The results also share simulations using *FIRESTAR3D* presently under development with good agreement with *FIRETEC* and *WFDS*. As stated earlier, these differences do not reflect on model accuracy, particularly due to inherent limitations of operational models (*MK5* and *BEHAVE*) under high wind speeds (>8 m/s) and the large spread in experimental data.

As a final point, we seek to illustrate with *FIRESTAR* how fuel

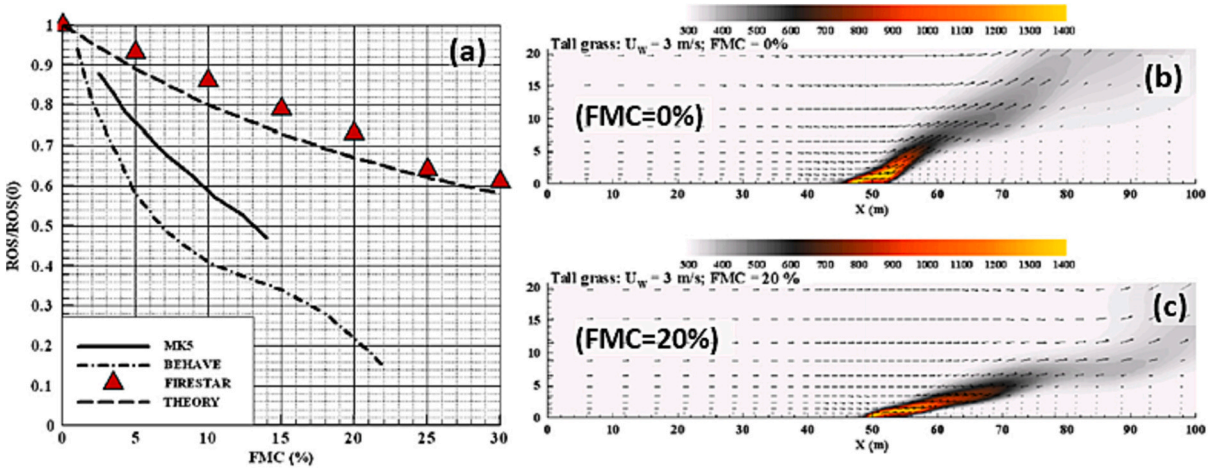


Fig. 17. (a) effects of fuel moisture content (FMC) on propagation rate of grassland fire as predicted by different models (ROS is the rate of spread, and ROS(0) is the rate of spread at 0% FMC); (b) fire line temperature field (indicated by color bar in K, and associated flame shape, y-axis is elevation in m) and velocity vectors for: (b) low (FMC 0%) and (c) high (20%) fuel moisture contents (From Morvan et al., 2009).

moisture content (FMC) affects ROS and flame geometry. *MK5* and *BEHAVE* show an exponential decrease in ROS with increasing FMC (Fig. 17a), whereas FMC effects predicted by *FIRESTAR* are moderate. FMC also affects flame geometry in the presence of wind as seen in Fig. 17b-c thus affecting radiative transfer of heat ahead of the fire line. Results show that fire lines intensity for grass with FMC = 0% was higher (~12 MW/m) than ~9 MW/m for grass with FMC = 20%. *FIRESTAR* offers many unique capabilities not found in other fully physical models; however, it is subject to similar limitations of unknown fuel heterogeneity, cutoff scales of process representation and other approximations (coupling between solid and fluid grids). As mentioned above, work is underway to extend *FIRESTAR* to three dimensions (Meradji et al., 2016).

8.4. CAWFE and WRF-Fire (Clark et al., 1996; Coen, 2005)

As the name implies, Coupled Atmosphere-Wildland Fire-Environment (*CAWFE*, Clark et al., 1996, 2004; Coen, 2005) model couples atmospheric flows with a wildfire model based on the local fire spread of Rothermel (1972) in similar form as used in *BEHAVE* model (Andrews, 2014). The model also includes the *BURNUP* algorithm (Albini and Reinhardt, 1995) for fuel consumption and secondary energy release after passage of the fire line. The projected coarse atmospheric grid is divided to surface fuel cells (30 m resolution) where the fire front location and unburned areas are tracked using four moving points (tracers forming a quadrilateral within a fuel cell) that represent the fire line interface. Tracers reaching cell boundaries can ignite neighboring cells if not already burning (provisions are made to avoid spurious ignitions). The model represents the flaming front positions, rate of spread (Rothermel, 1972), post-front heat release, and sensible and latent heat fluxes for surface and crown fires (Coen, 2013). The model can simulate small fires in the *LES* mode (fire eddies interacting in the atmospheric boundary layer), however, for large scale applications, *CAWFE* uses multi-meshing particularly near the fire line. The fluid dynamic model solves the Navier-Stokes and continuity equations for atmospheric flows using terrain following coordinates. The core of the atmospheric flows model interacts with the fire line behavior through sensible and latent heat fluxes. When a threshold heat flux is exceeded, fuel (canopy) ignites and the fire line within the cell is tracked as well as fire line between cells is advected consistently according to Richards (1995). Radiation is represented with an approximate solution. Refinements for considering spatially variable fuel density and type and diurnal variations in fuel moisture content have been implemented in *CAWFE* (Coen, 2005).

Coen (2013) and Mandel et al. (2011) have argued that numerical weather prediction (NWP) models can play a crucial role in realistic wildfire modeling due to their inherent physics and capability to represent background weather conditions and respond to fire perturbations at proper dynamics and scales. Hence, including wildfire physics into operational weather models such as the *Weather Research Forecast (WRF)* model (Mandel et al., 2011; Coen et al. 2013) was a logical evolution for harnessing knowledge and experiences gained in the development of *CAWFE*. *WRF* is an open-source model actively developed by a community of scientists thus benefiting from continuous improvements and is maintained by NCAR. It is portable and computationally efficient solver for conservation equations, fluid motion, atmospheric thermodynamics, and water states. The *WRF* model consists of various programs, including the *WRF* Preprocessing System (WPS), *WRF* Data Assimilation (*WRF-DA*), and for hydrologic applications, *WRF-Hydro* (Gochis et al., 2015). WPS is used to (i) define simulation domains, (ii) interpolate terrestrial data such as terrain and soil types, and (iii) convert, project, and interpolate meteorological data from other sources such as global models to the *WRF* simulation domain. *WRF-DA* can incorporate observations into the interpolated analyses created by WPS.

The *WRF-Fire* model uses a coarse atmospheric grid that contains a finer fuel grid, these two grids exchange information (see Fig. 18). The

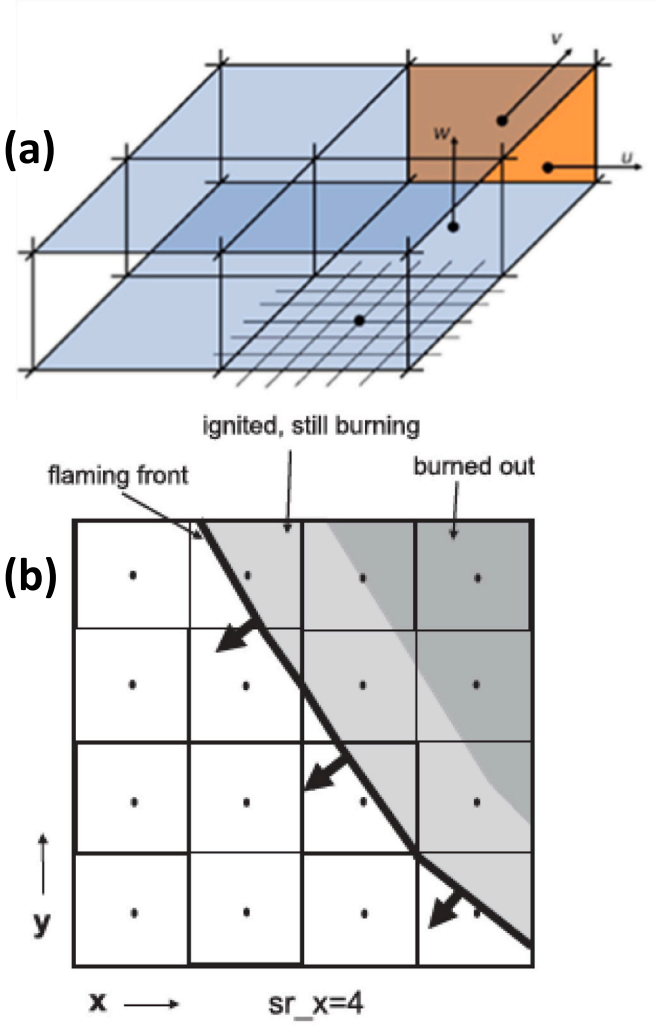


Fig. 18. WRF-Fire grid cells: (a) a 2 × 2 tile of the lowest atmospheric grid and the fuel surface mesh with wind vectors (u,v,w) located at midpoints of the atmospheric grid cell (Mandel et al., 2011); (b) surface fire mesh (fuel cells) with active fire line and burned-out regions (from Coen, 2013).

fire module is implemented as an added physics option in *WRF* where fire–atmosphere coupling occurs via wind, temperature, and moisture from the lowest *WRF* level to the fire module (applied at the fine fuel grid, often 30 m resolution). The fire module uses information to calculate fire spread and subsequent heat and water vapor emissions. The fire spread rate is based on Rothermel (1972) formulation with certain updates as discussed in Mandel et al. (2011). Heat release and fuel burned are calculated as a function of fuel type and elapsed time since ignition (similar to *CAWFE*, Clark et al., 2004). An important aspect of the fire line propagation is the use of level set formulation that solves a geometrical differential equation for the advance of the fire line with fire spread rates and ignition times specified at the center of fire grid cells. Water vapor and heat emissions (latent and sensible heat fluxes) from the fire are passed back to *WRF* and distributed vertically through an assumed extinction depth. *WRF-Fire* (Mandel et al., 2011) calculates heat fluxes determined from an exponential decay function and distributes the sensible heat flux vertically to account for radiation. Generally, *WRF-Fire* executes its simulation faster than real time even for highly resolved nested domains (the overall simulation time will depend on simulation domain, duration and computational resources). *WRF-Fire* was also coupled with *WRF-Chem* to trace the fire emission as the fluxes of carbon dioxide (CO2), carbon monoxide (CO), methane (CH4), and particulate matter (PM 2.5) released into the atmosphere at the location

of the fire. Coen (2013) stated that the convective-scale NWP models such as CAWFE have an advantage over WRF in being designed to simulate weather in a complex terrain and with steep variable topography gradients while *WRF* (a mesoscale model) aggressively dissipates fine-scale motions and gradients, distorting fire perimeter and strongly dampening fire signature due to internal smoothing.

8.5. QUIC-Fire (Linn et al., 2020)

QUIC-Fire is a unique wildfire simulation model that couples cellular automata (CA) rules for fire spreading with detailed modeling of fire-atmosphere physics (similar to FIRETEC). The CA component is inspired by Achtemeier's (2013) work and is guided by *QUIC-Fire* wildfire physical information that integrates the roles of wind-dominated fire spread with creeping and rate of transport of energy per unit time to derive CA spreading rules (Linn et al., 2020). While the fire spread module of *QUIC-Fire* is not fully physical, its use of physics-guided CA rules and detailed physics of fire behavior warrant its classification as a physical model. *QUIC-Fire* couples *QUIC-URB* a rapid wind field solver with *FIRE-CA* (the CA component) via fuel structure and fire effects on the evolving wind field and feeding back the spatially resolved vegetation and heat release to *QUIC-URB*. An advancing fire line alters the proximal wind field via buoyancy-driven flows (plume uplift and in draft of air to replace it) and alteration of drag as vegetation is consumed and the landscape changes. The *QUIC-URB* is adjusted to consider wildfire effects and feedback. The *FIRE-CA* component of *QUIC-Fire* considers at each cell reaction, mass loss and heat release rates, that, in turn, generate a batch of discrete energy packages (EP) that are subsequently distributed to neighboring cells. When EPs reach their destination, based on current conditions in the cell, EPs can heat up the fuel and evaporate moisture or ignite the fuel, contribute to intensifying existing fire or transfer energy to the atmosphere. The distances and modes of EPs transport are determined by wind conditions or creeping via surface fuel consumption. Critical parameterization schemes required for establishing the CA in *QUIC-Fire* were guided by experimental data or by high fidelity *FIRETEC* simulation results. *QUIC-Fire* was designed for prescribed fire planners to generate scenarios and address complex atmospheric feedback faster than real time simulations. *QUIC-Fire* runs nearly 3 orders of magnitude faster than CFD-enabled models such as FIRETEC (Linn et al., 2020).

9. Summary of wildfire physics and models

The primary aim of reviewing wildfire basic physics, introducing historical wildfire model applications and a brief overview of contemporary wildfire simulation models was to provide hydrologists, ecologists and soil scientists with easy access to wildfire science, the present state of practice in modeling and representing this important natural phenomenon. We considered three broad types of wildfire models: (i) statistical-empirical, (ii) semi-physical and (iii) fully physical-mechanistic. However, our primary focus was on operational models that represent the complete range of wildfire behavior and fire line propagation including related processes of burnout (and fire spotting) that play a critical role in the lasting impacts of wildfires on landscapes. The interested reader can also identify the different software packages associated with the various wildfire models. Developing a clearer understanding of how wildfires spread, links to fuel (discussed next), landscape topography and weather and related metrics such as radiative emissions from fire lines are prerequisites for the next challenge of systematically evaluating wildfire effects on soil properties and thermal alterations that affect soil functioning as discussed below.

9.1. The role of fuel distribution and characteristics in wildfire behavior

Wildfire spread and behavior are sensitive to the spatial distribution and characteristics of fuel (i.e., the flammable land cover essential for fire spread – Duane et al., 2021). The dependency of wildfire spread on fuel structure and characteristics was recognized as an important factor since the early days of wildfire modeling (Rothermel, 1972; Albini, 1976; Anderson, 1982). Nevertheless, the systematic translation of how spatial heterogeneity in fuel type, loading and other traits affect wildfire advance and behavior remains a challenge. Studies of wildfire response to different arrangements of fuel have relied primarily on ensemble model simulations that keep all other variable constant except fuel arrangement (Finney, 2006; Duane et al., 2021; Aparício et al., 2022; Atchley et al., 2021). A recent systematic study of Atchley et al. (2021) employed the mechanistic *FIRETEC* model to illustrate that wildfire spread is typically hampered by the presence of gaps and other fuel heterogeneities (relative to spread across uniform fuel distribution). However, beyond a certain spatial correlation length of fuel heterogeneity, wildfire spread increased again due to interactions among continuous fuel features and the resulting local airflows and turbulence. Aparício et al. (2022) analyzed how topological aspects of fuel connectivity affect fire percolation thresholds in which wind plays an important role in the resulting fire spreading pathways. The directionality and rate of spread are jointly shaped by prevailing winds and fuel arrangement. These aspects have been studied by Finney (2006) in the context of minimum travel time (*MTT*) that provides a useful index of fuel connectivity and helps landscape managers design fuel breaks and plan fire containment scenarios (Aparício et al., 2022). The *MTT* calculates wildfire growth by identifying the paths with the minimum fire line travel time among nodes of grid (using a variant of FIRESITE called *FlamMap* - https://www.firelab.org/project/flammap). Aparício et al. (2022) expanded on the work of Finney (2006) by using *FlamMap* to derive landscape connectivity indices for wildfires under different weather scenarios. The results are important for identifying critical locations for placing fuel breaks to contain or redirect wildfires and potential locations for effective suppression efforts (i.e., regions with low fire line intensity). This sample of recent studies focusing on linking fuel structure (and composition) with wildfire behavior also provides information on the distribution of fire line intensities (or local fire radiative energy, FRE) a critical component for assessing wildfire impacts on soil over landscapes to be discussed shortly.

Early efforts to classify vegetation type, density and structure into parameters used by models (i.e., a fuel model system) were part of the National Fire Danger Rating System (*NFDRS*) that developed metrics for fire danger maps for a range of conditions. The 1972 NFDRS system used 20 different fuel models based on vegetation type, load and state. The concept of Fire Behavior Fuel Model (*FBFM*) was modified by Anderson (1982) to include 13 fuel types in four groups (grass, shrub, timber, and slash). Because fuel classification systems were driven by the requirements of specific wildfire models, they are not general, and a selection of a fuel class may be manifested in different ways in model performance (Price and Germino, 2022). For example, one fuel class may accurately represent the rate of wildfire spread whereas a different class may be needed to properly represent fire intensity (Anderson, 1982). Scott and Burgan (2005) have expanded the fuel classes for the Rothermel (1972) fire spread model to define 40 fuel variants or models including dynamic aspects of greening-up or grass curing. A recent study by Price and Germino (2022) have used the FARSITE model to illustrate the sensitivity of model performance to specifics of assigned fuel model, for example, using readily available maps provided by the *Landfire* initiative (https://www.landfire.gov/index.php). The study by Price and Germino (2022) highlights the importance and challenges of constructing reliable FBFMs in the resulting model predictions of shape and extent of burned areas. Without delving into technical details of how FBFMs maps are constructed, the primary challenge is to establish for every pixel of the landscape a representative vegetation composition and fraction of different vegetation groups, their areal density, and other features of fuel arrangement (i.e., the position of branches and other large fuel elements on the ground or slash left after clear-cut timber harvest). Contemporary fuel classification systems (Ottmar et al., 2007)

attempt to generalize fuel representation via Fuel Characteristic Classification Systems (FCCS) that address these challenges by adding vertical information (fuel orientation), dynamic fuel states and other criteria. Ottmar et al. (2007) developed an FCCS focused on the US, however the framework is general and can be applied worldwide. Remote sensing is playing an important role in developing modern fuel loads and fuel type maps (Arroyo et al., 2008), for example, the development of updatable spatial fuel data (maps) for the *Landfire* initiative in the US (Reeves et al., 2009, see next section for additional information) and similar initiatives in Europe (Baetens et al., 2022).

9.2. Regional and global georeferenced spatial and temporal datasets for wildfire modeling

As presented in the previous section, some of the most critical inputs for the application of wildfire models at the landscape scale are fuel, weather and topographic information. The rapid proliferation of advanced remote sensing platforms for vegetation cover, land disturbances, direct wildfire monitoring (NASA LANCE/FIRMS https://firms.modaps.eosdis.nasa.gov/), landscape topography and highly resolved climate data offer unprecedented opportunities for wildfire modeling at high resolution across many regions of the world. Many of the databases are periodically updated such as maps of fuel and weather data. The rapid advances in machine learning for parameter estimation and pattern extraction (delineating wildfire boundaries and dynamics) play an ever-increasing role in model evaluation and creation of global databases of wildfire related geospatial information. In the US and territories, the *Landfire* initiative represents one of the most extensive wildfire-related database with over 25 geospatial layers of updatable variables grouped into georeferencing, disturbances and changes in landscapes, vegetation types and traits, fuel types, historical fire regimes, topography and seasonal (temporal) information. The native spatial resolution of *Landfire* products is 30 m and dynamic variables are now updated annually. *Landfire* is a joint endeavor of the Forest Service of the US Department of Agriculture and the US Department of the Interior operating under the *Wildland Fire Leadership Council* (*WFLC*). A related product is the *Wildfire Hazard Potential* (*WHP*) map developed by the Fire Modeling Institute of the US Forest Service (Dillon et al., 2015). The *WHP* maps are informed by *Landfire* data coupled with *Large Fire Simulator* (*FSim*) for deriving the *WHP* index at 270 m resolution across the US. The primary purpose is to delineate regions in a need for vegetation (fuel) management towards reducing the intensity of future fires and not a forecast for future fires. Similar capabilities are also offered by private entities such as Esri (https://www.esri.com/en-us/industries/wildland-fire/strategies/fuel-management). Similar initiatives have been developed globally such as reported by Zhang et al. (2021) and by Shmuel and Heifetz (2022) on application of machine learning and deep neural networks for global wildfire susceptibility modelling.

Globally, several databases for wildfire modeling similar to *Landfire* (vegetation cover, digital elevation, soil type maps and weather information) are presently available such as the *Global Wildfire Information System* (GWIS) developed by the European Union (https://gwis.jrc.ec.europa.eu/) building upon ongoing activities of the *European Forest Fire Information System* (EFFIS, https://effis.jrc.ec.europa.eu/). For example, Artés et al. (2019) have developed a global wildfire dataset as part of GWIS. It is intended for analyses of fire regimes and behavior. This is meant to bridge the gap between remote sensing information and mechanistic wildfire behavior (providing information on fire size, fire spread speed, how fires may evolve and merge into a single event, and the number of fire events for a given area). Other initiatives have been advancing the integration of wildfire modeling and society such as the Leverhulme Centre for Wildfires, Environment and Society in the UK (https://centreforwildfires.org/projects/) or FLARE Wildfire Research group based in the School of Ecosystem and Forest Sciences at The University of Melbourne (https://www.flarewildfire.com/) that attempt to harmonize information and modeling to better prepare societies to cope with wildfires. The expansion of earth observing capabilities and concerted efforts by various agencies to develop new databases and make these widely available for wildfire modelers and for model testing offer unprecedented opportunities for linking wildfire science with ecology, hydrology and soil science. The wealth of information permits in-depth analyses of wildfire affected landscapes and their recovery over time.

9.3. Looking down – wildfire effects on soil and landscape processes

Traditionally, wildfire modeling and study have focused on fire safety engineering, wildfire control strategies, as well as smoke and particle emissions to the atmosphere. The primary motivation of this review of wildfire physical processes and modeling representation was the need for a quantitative (and ultimately predictive) framework for assessing impacts of wildfires on soil functions and hydrologic process. Wildfires affect large tracts of natural lands with significant alterations to ecological and hydrological processes. The pathways for post-fire landscape recovery depend on the degree of soil thermal alterations, local climate and on various biophysical processes that operate across different spatial and temporal scales (years to centuries). For example, depending on fire temperature and duration (reflected in burn severity), surface soil organic carbon degrades and becomes pyrolyzed or undergoes combustion that, in turn, may result in loss of soil structure affecting soil infiltration capacity and erodibility with enhanced runoff and potential for flooding, higher rates of soil erosion and potential for hazardous debris flows (McGuire et al., 2017; Alessio et al., 2021). Other thermal alterations include condensation of volatiles formed during pyrolysis (within the soil) that may render soil surfaces hydrophobic and further reducing infiltration capacity of the landscape (DeBano, 2000). At a catchment scale, the magnitudes of soil surface alterations, their spatial distributions, and the duration of these changes play a critical role in the hydro-ecological functioning of burned catchments and their rates of post-fire recovery (Williams et al., 2022).

Remarkably, important effects of wildfires on soil properties and processes remain poorly represented in most modern wildfire models. In part, due to the focus on atmospheric processes critical for the representation of fire behavior and spread, most likely, the primary reason for the glaring omission of wildfire effects on soils reflects the complexity of the problem. Even the seemingly simple task of translating the passage of a fire front with known fire radiative energy (FRE) density to a soil heat flux or surface temperature requires numerous assumptions and calculations (Linn and Cunningham, 2005). The incident and net radiative heat flux received on the soil surface can vary widely with local fire line characteristics (heat release rate, flame height, and viewing factors) thus presenting a challenge to reliable estimation of net heat fluxes at the soil surface necessary for solving the heat transport into the soil profile. Even with advanced numerical models, this is not a trivial task. Moreover, even with the soil heat transport resolved, the translation of thermal perturbations to changes in soil physico-chemical properties and expressing these in terms of measurable physical (soil structure and hydrophobicity), hydrologic, and ecological parameters remain an active topic of research. In the following, we review simple approaches for translating wildfire characteristics and associated information to soil thermal boundary conditions and heat transport and highlight knowledge gaps in translation of wildfire effects on quantitative thermal alteration of soil processes.

10. Soil surface and subsurface thermal alterations

10.1. Empirical Burn Severity Indices

In the absence of a landscape scale quantitative framework for translating fire characteristics to spatially (and temporally) resolved soil-altering thermal regimes, the community has relied on empirical visual and remotely sensed surrogates that provide indices such as soil

burn severity (Parson et al., 2010; Keeley, 2009; Parks et al., 2014; Smith et al., 2016). The rationale behind these classification methods is that different wildfire intensities (convective and radiative energy release rates) and associated temporal temperature fields imprint distinct signatures on fire-affected landscapes, and these signatures can be remotely sensed via spectral measurements as well as visually classified using calibrated standards and cues at the landscape scale. In a review of methods and nomenclature, Keeley (2009) highlighted the importance of distinguishing between metrics related to fire intensity (i.e., local energy release) and fire consequences in terms of burn severity (of vegetation or soil). High and short-lived fire intensity such as crown fires in forest or rapidly advancing grassland fires may have only minor effects on soil burn severity (Parson et al., 2010). Smith et al. (2016) criticized ad-hoc approaches for burn severity classification that sometimes confuse correlation with causation; the authors proposed a more systematic experimental approach for quantifying burn severity based

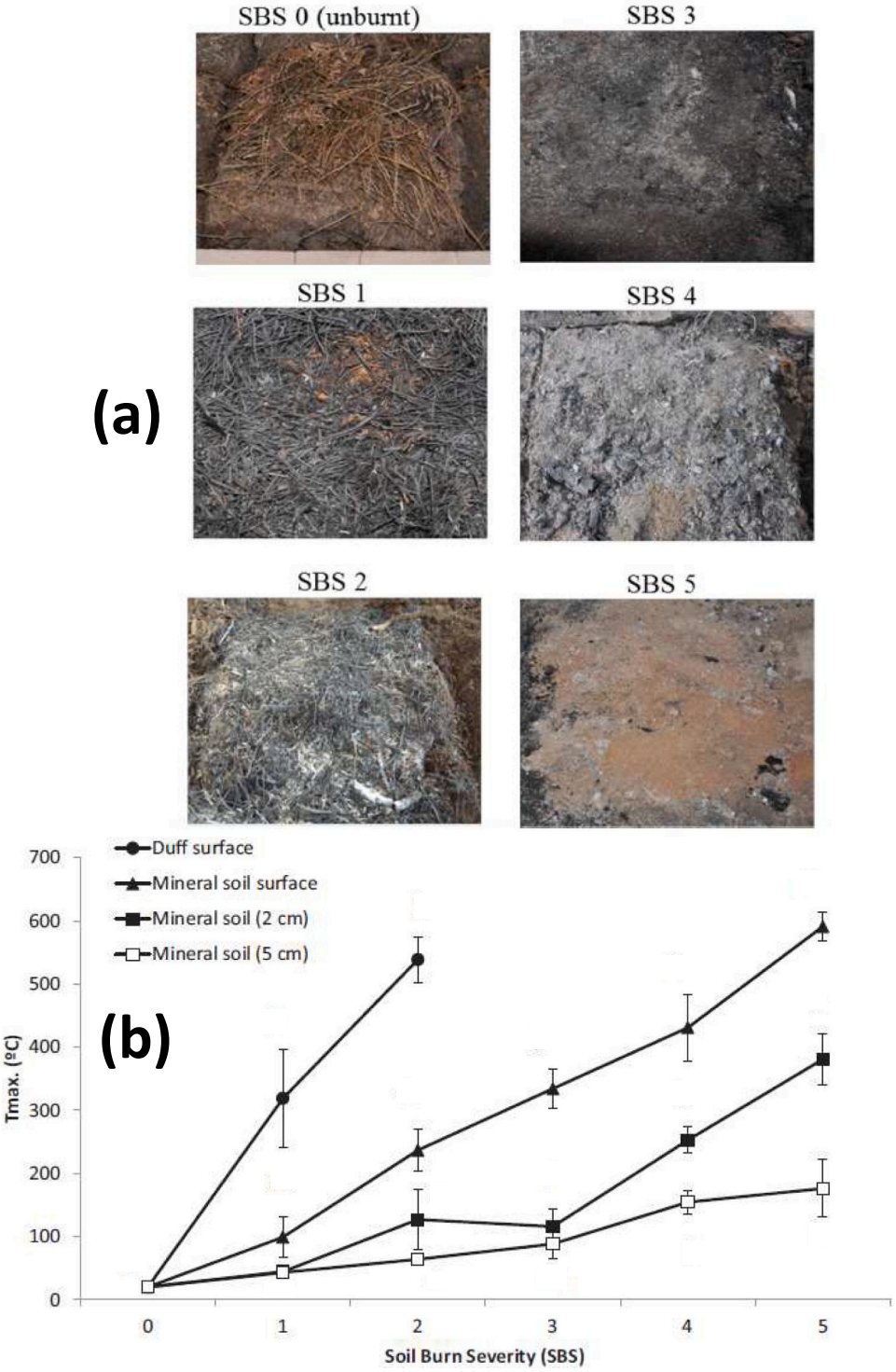


Fig. 19. (a) Illustrative examples of soil burn severity (SBS) classes with partial (left) and complete consumption (right) of the soil organic layer by the fire; (b) maximum temperatures experienced at the duff and soil surfaces associated with different soil burn severity classes. Note that for SBS > 2, all the organic layer was consumed (from Merino et al., 2018).

on direct and well- controlled small-scale burn measurements. While such an experimental approach alleviates shortcomings of the empirical approach, its applicability to wildfire response at (heterogenous) landscape scales remains to be seen. Often, burn severity indices reflect loss or changes in organic matter above or belowground (i.e., soil organic matter, SOM) that, in turn, alter ecosystem behavior (hydrological and ecological). The practical estimation of burn severity indices is often based on a field guide by Parson et al. (2010) that summarizes the state of practice and terminology used by Burned Area Emergency Response (BAER) teams to quantify post-fire soil burn severity (including metrics for vegetation and reflectance classification). While the rapid post-fire assessment and maps produced by the BAER teams are critical for risk assessment and rehabilitation efforts, they suffer from lack of consistency due to the empirical nature of the indices. Some of these challenges have been alleviated with the availability of dedicated remote sensing products for pre- and post-fire classification (Clark and Bobbe, 2006) to assist with BAER teams' assessment. Briefly, these products are based on the Normalized Burn Ratio NBR that identifies fire effects on vegetation and landscapes based on differences in near infrared band (NIR) and the short-wave infrared band (SWIR) that respond to fire-induced vegetation alterations. A derived product called the differenced normalized burn ratio (dNBR) uses differences in pre- and post-fire NBR and a suite of similar scaled indices have been shown to reduce misclassification (Parks et al., 2014). We note here that inferences based on NBR have been challenged (Roy et al., 2006) and are of limited value in semiarid landscapes. In an experimental study by Merino et al. (2018), intact top soil monoliths with organic horizons and duff were burned in a combustion wind tunnel to simulate a range of fire conditions enabling measurement of changes in SOM as a function of surface temperature were reasonably well correlated with maximum temperatures measured at the different surfaces (see Fig. 19) supporting the utility of empirical classifications schemes such as delineated in Parson et al., 2010) for spatially extensive post-fire classification (such as performed by BAER teams).

10.2. Towards a quantitative framework for soil burn severity estimation

Most modern wildfire models stop short of calculating heat fluxes intercepted by soil surfaces. Consequently, impacts of wildfire models on soil thermal alteration require post-processing of fire metrics to estimate duration, fire radiative energy fluxes and temperatures at the soil surface towards affecting changes in soil organic carbon and related properties (Freeborn et al., 2008). While most processes and parameters for performing these computations are readily available from wildfire modelling platforms combined with adaptation of models from soil science (Campbell et al., 1995; Albini et al., 1996; Enninful and Torvi, 2008; Massman, 2012; Massman, 2021) and from fire safety engineering (Zhai et al., 2019; Gong et al., 2020b; Parot et al., 2022), yet to date, no studies have applied these quantitative approaches to fire-affected soils at the catchment scale. Admittedly, this is not a trivial task, and various challenges hinder the extension of modern modeling capabilities including: (1) translation of information from the dynamic and highly transient fire line to incident radiative fluxes at the soil surface that require estimation of flame geometry and angle, view factors and radiative surface energy balance that are all poorly constrained; (2) soil water content and phase changes in the soil complicates heat conduction (Scotter, 1970; Massman, 2021); (3) coarse resolution and spatial averaging in landscape scale wildfire models (e.g., WRF-Fire) limit information required for solving local soil heat transport.

Addressing these modeling challenges is beyond the scope of the review, yet a few promising simplifications may offer a way towards quantitative modeling of thermal effects of wildfires on soil properties. One approach is to harness information from the wildfire model regarding characteristics of the fire line (mean ROS, energy release and ambient temperatures) towards applying simple diagnostic calculations in a post processing mode. For quantitative screening of wildfire surface temperature on soil surfaces and the depth of the impacted soil processes and properties, a simplification of the complicated surface temperature dynamics during the wildfire may be needed. For example, it is simple to use the one-dimensional heat equation considering conduction only with prescribed initial conditions (ambient temperature) followed by an instantaneous jump in air temperature caused by a fire. The 1D heat conduction equation is:

$$\frac{\partial T}{\partial t} = D\frac{\partial^2 T}{\partial x^2} \tag{12}$$

Where T is temperature, t is time and D is the thermal diffusivity (L^2/t). The thermal diffusivity is a soil property that may vary with soil texture, water content and soil organic matter content, and is defined as $D = k/(\rho C)$ where k is the bulk soil thermal conductivity (ML/t^3 K or in MKS units: J $m^{-1}s^{-1}K^{-1}$)), ρ is the soil bulk density (ML^{-3}), and C is the specific heat per mass ($L^2/t^2/K$ or J kg^{-1} K^{-1}). The soil surface is set as $x = 0$, with x increasing (positive) downwards into the soil.

First, we consider the simple case of a step change of the soil surface temperature (mimicking instantaneous arrival of the fire with known initial and boundary conditions) with an initial (equilibrium) temperature prior to start of fire denoted by: $T = T_{eq}$ assuming that with the arrival of the fire line, the air temperature instantaneously reaches T_0 ($t \geq 0$, $T = T_0$), and the soil temperature then evolves as a function of depth and time ($t \geq 0$, $T = f(x,t)$). These initial and boundary conditions are given as:

$$\begin{aligned} T(x,0) &= T_{eq}\ (IC) \\ T(0,t) &= T_0\ (Top\ BC) \\ T(\infty,t) &= T_{eq}\ (Bottom\ BC) \end{aligned} \tag{13}$$

An analytical solution for the 1-D heat conduction for boundary conditions of Eq. (13) is (Carslaw and Jaeger, 1959):

$$\frac{T - T_{eq}}{T_0 - T_{eq}} = erfc\left[\frac{x}{\sqrt{4Dt}}\right] \tag{14}$$

Where *erfc(u)* is the *complementary error function* available in most mathematical packages (including MS Excel). Soil thermal properties (particularly thermal diffusivity) can be obtained from literature values (e.g., Table 5). For illustrative purposes, we present in Fig. 20 analytical predictions of soil temperature profiles resulting from a constant elevated temperature (750^0 C) over dry (full lines) and wet (dashed lines) silt loam soil surface. These simple thermal boundary conditions may develop under slash pile burns that may persist over many hours as described by Campbell et al. (1995) and Massman (2012). This simplified calculation assumes uniform soil properties and does not consider

Table 5
Thermal properties of soil constituents and natural soils.

Soil constituents	Thermal Conductivity (k) [W/m-K°]	Density (ρ) [kg/m³]	Specific Heat (C) [J/kg-K°]	Thermal Diffusivity (D) [m²/s]
Quartz	8.80	2660	750	4.40E-06
Clay Minerals	3.00	2650	760	1.50E-06
Soil Organic Matter (SOM)	0.30	1300	1900	1.20E-07
Water	0.57	1000	4180	1.40E-07
Air	0.025	1.293	1006	1.85E-05
Soil textural class				
Sand (dry)	0.30	1400	750	2.8E-07
Sand (wet - max)	1.40	1600	1300	7.7E-07
Silt Loam (dry)	0.22	1250	850	2.0E-07
Silt Loam (wet - max)	0.90	1450	1150	5.2E-07

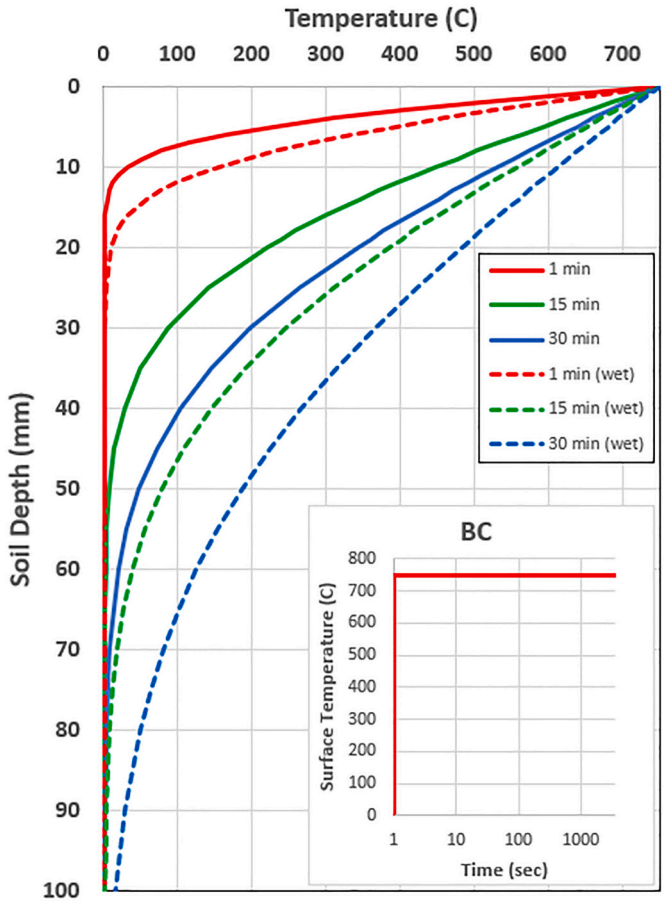


Fig. 20. Soil temperature as a function of depth from a constant heat source at the soil surface (750^0 C) with thermal diffusivity for dry silt loam soil 0.2 mm2/s (full lines) and wet soil with 0.5 mm2/s (dashed lines) at three elapsed times 1, 15 and 30 min since the start of fire (or arrival of fire front).

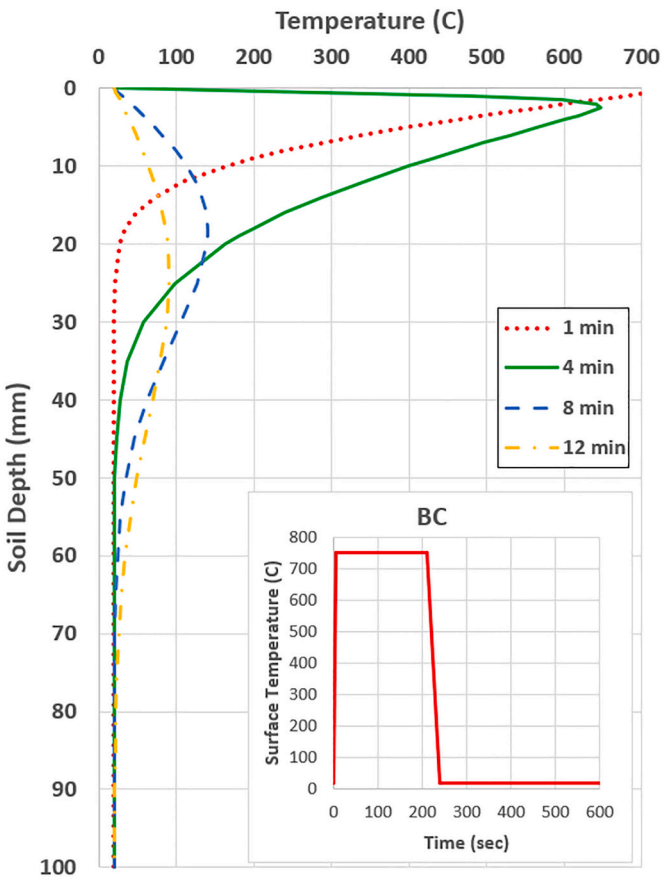


Fig. 21. Soil temperature as a function of depth due to passage of a fire front with maximum soil surface temperature of 750^0 C that lasted for 238 s (see inset for boundary conditions) over a silt loam soil with assumed thermal diffusivity of 0.5 mm^2/s. Soil temperature profiles for 1, 4, 8 and 12 min since arrival of the fire front.

important effects of soil moisture that affect soil thermal properties (as with soil thermal diffusivity in the example), but also phase change (energy consumed by water changing from liquid to vapor) within the soil that affects or even "arrests" the propagation of the heat front until the soil dries (Scotter, 1970; Albini et al., 1996; Badía et al., 2017). The challenges of modeling interactions between soil moisture dynamics and thermal regimes under wildfires have been reviewed by Massman (2021).

Next, we consider more realistic boundary conditions imposed by rapid passage of a fire line with residence times varying with fuel type and atmospheric conditions (1–3 min for grassland fires to 10 min for forest fires with sparse surface fuel). This is represented by the passage of high surface temperature T_0 that persists for t_p over a surface (followed by restoration of ambient air temperature at the surface):

$$\begin{gathered} T(x,0) = T_{eq}\ (IC) \\ T(0,t) = T_0\ for\ 0 \le t \le t_p\ (Top\ BC) \\ T(0,t) = T_{eq}\ for\ t > t_p\ (Top\ BC) \\ T(\infty,t) = T_{eq}\ (Bottom\ BC) \end{gathered} \tag{15}$$

The analytical solution for the 1-D heat conduction (Eq. 12) subject to the conditions in Eq. 15 is given by (Carslaw and Jaeger, 1959; Scotter, 1970):

$$\begin{gathered} \frac{T - T_{eq}}{T_0 - T_{eq}} = erfc\left[\frac{x}{\sqrt{4Dt}}\right]\ for\ t \le t_p \\ \frac{T - T_{eq}}{T_0 - T_{eq}} = \left\{ erf\left[\frac{x}{\sqrt{4D(t - t_p)}}\right] - erf\left[\frac{x}{\sqrt{4Dt}}\right] \right\}\ for\ t > t_p \end{gathered} \tag{16}$$

Where *erf(u)* is the *error function*. In contrast with the constant change in surface temperature case, the results depicted in Fig. 21 demonstrates the formation of a heat wave that simultaneously travels and decays with time and depth. The results depicted in Fig. 21 reflect a maximum surface temperature of 750^0 C that lasts 4 min with thermal perturbation limited to shallow depths where temperature values exceeding 100^0 C rarely penetrate below 30 to 50 mm (Badía et al., 2017).

For real wildfires, a fire line imposes a variable radiative heat flux boundary condition that must enter numerical or analytical solutions of the heat conduction into the soil (Enninful and Torvi, 2008; Parot et al., 2022). This top boundary condition considers the incident radiative flux on the soil surface (q_{in}, Wm^{-2}) balanced by convective and radiative losses (*reradiation*) from the rapidly heated soil surfaces (with coefficients $h = h_C + h_R$ [W m^{-2} K^{-1}]) that jointly determine a conductive heat flux into the soil (k is soil thermal conductivity):

$$-k\frac{\partial T}{\partial X}\bigg|_{x=0} = q_{in} - h(T - T_{eq})\ (Top\ BC) \tag{17}$$

For simplicity, a linearization of the outgoing radiative heat flux emitted from the heated soil surface (reradiation) reduces this nonlinear

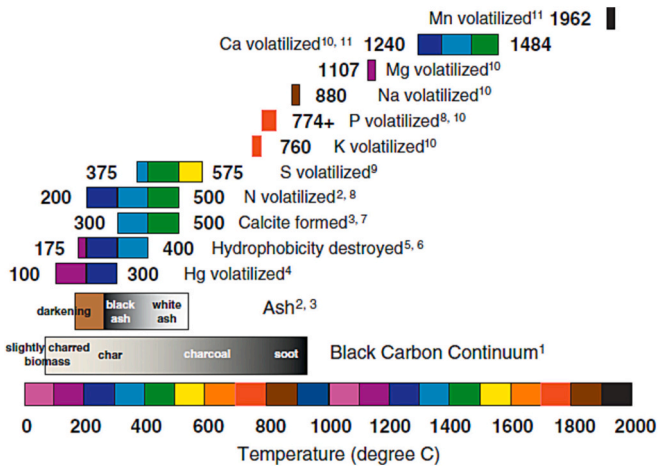


Fig. 22. Temperature thresholds and ranges for thermal alterations of soil constituents and ash (from Bodí et al., 2014).

term into a form similar to convective heat losses (proportional to surface-air temperature gradient) with the coefficient $h_R = \epsilon\sigma T_{avg}^3$ (with ε the emissivity; σ Stefan-Boltzmann constant 5.67×10^{-8} W m^{-2} K^{-4} and T_{avg} is the mean surface temperature; see Parot et al., 2022 for details). Various forms of time variable radiative boundary conditions have been proposed in the literature including exponentially increasing surface heat flux (Gong et al., 2020a); linearly decaying surface heat flux (Zhai et al., 2019; Gong et al., 2020b) and other radiative boundary conditions that can be adapted to wildfires as reviewed by Parot et al. (2022). Despite limitations of the diagnostic computations presented above, they provide consistent approximations for assessing potential depth of alteration induced by different temperatures. Across a wide range of wildfire dynamics and resident times (t_R = flame width/ROS), these calculations confirm trends known from field observations (Albini et al., 1996; Stoof et al., 2011b) that wildfire thermal alterations tend to concentrate near the top 30–50 mm of the soil profile (Steward et al., 1990). For certain conditions, such as under slash pile burns (Massman, 2021) or for prolonged smoldering and burnout of large fuel elements (Albini and Reinhardt, 1995; Burrows, 2001), soils may be affected to depths of hundreds of mm (e.g., Fig. 20). Ulery et al. (1996) estimated the relative area of deep thermal alterations to represent 1–2% of the burned area.

10.3. Thermal alterations of soil chemical and physical properties

A quantitative representation of wildfire effects on soil would enable the harnessing of knowledge from soil chemistry, combustion science and soil physics (Araya et al., 2016; Araya et al., 2017; Lautenberger and Fernandez-Pello, 2009; Parot et al., 2022) to delineate chemical and physical alterations of soil profiles (Araya et al., 2017; Ulery et al., 1996; Certini, 2005) towards process-based and spatially resolved predictions of changes in soil properties that, in turn, affect changes in hydrology, ecology, debris flows or landscape recovery rates. Numerous studies have linked the onset of specific temperature thresholds with chemical alteration of soil constituents (Figs. 22 and 23), such alterations are not limited to SOM only, but at sufficiently high temperatures, the mineralogy of the soil itself changes with volatilization of major cations (Ulery et al., 1996; Araya et al., 2016; Bodí et al., 2014) thus irreversibly affecting soil composition and post-fire element cycling (Roshan and Biswas, 2023).

Arguably the most important wildfire-induced landscape and soil alterations are related to changes in organic carbon, from complete combustion of above ground canopies to partial pyrolysis of soil organic carbon (SOC) (González-Pérez et al., 2004). Wildfire induced above ground alterations of forest and grassland canopies and vegetation is probably the most visible and recognizable effect of wildfires. Less distinct and equally important for landscape functioning is the loss of the organic binding agents that hold soil particles together to form aggregates and other soil structure features. The alteration and even loss of soil structure at the upper most layer of the soil where rainfall and gases enter or leave the soil is crucial for rainfall partitioning to infiltration and runoff, that affect recharge and erosion rates. Studies have shown how SOC carbon is gradually lost with increasing soil heating (typically at temperatures above 100^0 C) as illustrated in Fig. 23 a (Araya et al., 2017). In this figure, the authors also show loss of soil organic nitrogen with temperature (Fig. 23 b – note the higher temperature threshold of 250^0 C for loss of organic N relative to SOC). Associated with SOC losses due to soil heating are significant changes in soil structure as illustrated in Fig. 24 (Massman et al., 2010) that support other quantitative measurements of loss of soil aggregation (Certini, 2005; Araya et al., 2016; Sion et al., 2023) that result in reduced mechanical stability and enhanced soil erodibility and potential for debris flows (Alessio et al., 2021). While evidence of enhanced soil hydrophobicity following wildfire (DeBano, 2000; Stoof et al., 2011a) have received considerable attention due to their direct links to soil hydraulic properties and

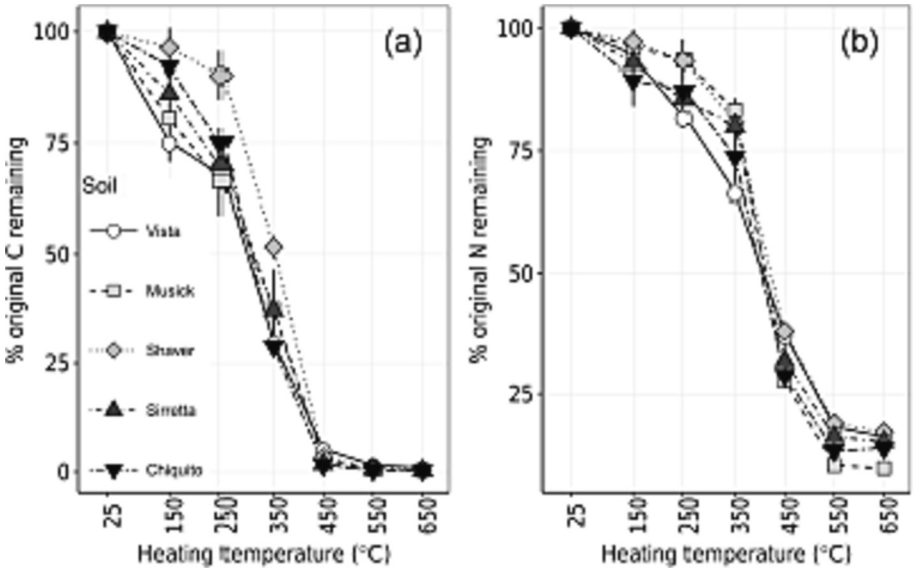


Fig. 23. Fraction of (a) soil organic carbon and (b) organic nitrogen loss from different soils as a function of heating temperature (from Araya et al., 2017).

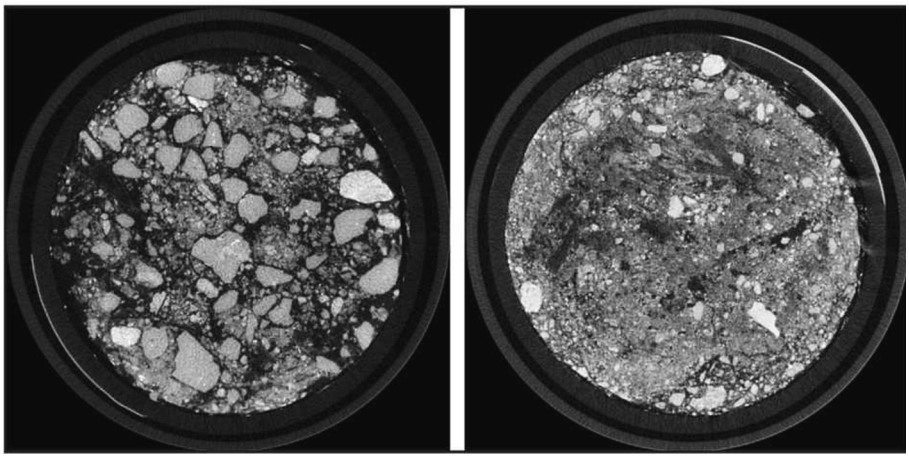


Fig. 24. Effects of slash-pile burn on soil structure: (left) X-ray tomography of natural soil sample (10 mm depth; 36 mm in diameter) not affected by fire; (right) a soil sample at the same depth showing fire effects (from Massman et al., 2010).

infiltration behavior (Stoof et al., 2012; Wieting et al., 2017; Shillito et al., 2020), we estimate the long-term wildfire impacts on landscape response and recovery rates to be strongly linked to loss of soil structure and other depth-dependent physical, chemical and ecological thermal alterations.

In a review of effects of prescribed fires on soil properties, Alcaniz et al. (2018) argue that for low intensity fires with temperatures below 220^0 C, no significant effects on soil aggregate stability have been observed. However, because oxidative pyrolysis of soil organic matter (SOM) is a gradual process that requires threshold temperatures (Araya et al., 2017, Fig. 23a) and a certain pyrolysis time (1–3 min, González-Pérez et al., 2004), it is reasonable to expect structural changes whenever soil temperatures exceed 100^0 C and last a few minutes (e.g., at the top 20–30 mm of the soil in the hypothetical example in Fig. 21; and as shown by Albalasmeh et al., 2013 and Jian et al., 2018 for low intensity fires).

A prominent effect of wildfires on soil ecology is via changes in vegetation cover and primary production (including damage to roots) that alter carbon inputs to soil and adversely impact soil respiration and microbial activity. Several studies have investigated the effects of wildfires on soil microbiome diversity and function, for example, Nelson et al. (2022) report significant effects of wildfires on soil microbiome diversity and function. Likewise, Holden et al. (2015) measured a reduction in soil respiration due to indirect effects on soil water content (drier conditions) and changes in the quality of soil organic compounds available soil microbes. Some of the direct effects of wildfires on soil ecology is attributed to lethal temperatures for life in soil that has been considered as exposure to temperature > 60 °C for over a minute. Pingree and Kobziar (2019) have critically evaluated this common threshold and concluded that it is an overly simplified metric that overlooks soil heterogeneity and the wide range of physiological traits of soil biota (most organisms show more resilience to this prescribed threshold, and some organisms avoid the threshold by migrating deeper into the soil). Fig. 25 provides a visual summary of how wildfires induce differential thermal alterations with soil depth that, in turn, alter pre-fire soil structure, induce hydrophobicity, modify SOC via pyrolysis, alter

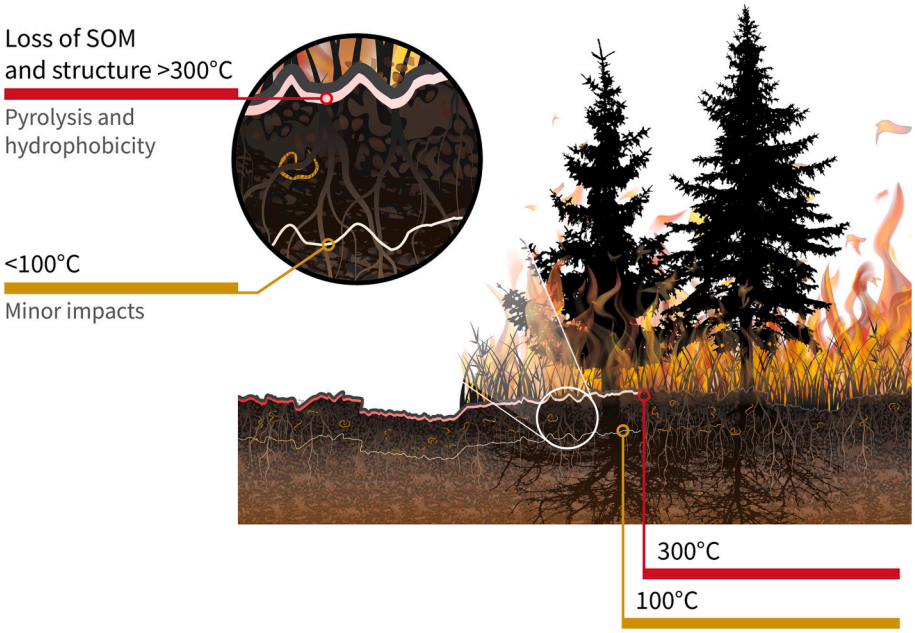


Fig. 25. Conceptual links between wildfire induced thermal alterations and changes in soil properties. Different critical temperatures and their penetration depths will alter soil structure, induce hydrophobicity, modify SOM/SOC via pyrolysis, alter soil minerals, and impact soil fauna and ecological functioning to different degrees. Hence the importance of establishing direct links between wildfire characteristics and the spatial distributions of critical soil temperatures with depth.

existing mineralogy and impact soil fauna and ecological functioning. The degree of these alterations and their spatial distributions play an important role in the resulting post-fire landscape hydro-ecological functioning and the rates and pathways for recovery discussed next.

10.4. Wildfires hydro-ecological impacts and landscape recovery

Fire-induced alterations to soil ecology and hydrology have been studied extensively and summarized in books on the subject such DeBano et al. (1998) dedicated to fire effects on soil and water resources; Keeley et al. (2012) monograph that summarizes knowledge on wildfire effects in Mediterranean ecosystems; and a recent global literature analysis by Roces-Díaz et al. (2022) for fire effects on ecosystem services of forested landscapes. While detailed description of these effects is beyond the scope of this review that focuses on the links between wildfire dynamics and physical processes and soil alterations, we seek to highlight here a few principles that link the magnitude of the wildfire perturbation to soil and hydrologic response and list characteristic time scales for post-fire landscape recovery (i.e., the persistence of the disturbance). The loss of vegetation and destruction of near-surface soil structure reverberate throughout the landsacape's modified hydrology in the years following the wildfire (e.g., more runoff with enhanced erosion and soil loss, less infiltration and evapotranspiration). In an influential paper, Shakesby and Doerr (2006) summarized the many changes in hydrology and sediment yields across many catchments with nearly 2 orders of magnitude increase in average soil loss for the first three years following the wildfire. The absolute values scale with the burn severity, local climate, topography, and vegetation. Mayor et al. (2007) have reported significant increases in runoff and sediment yields that persist several years after the wildfire event. For Mediterranean climate with mean annual precipitation of 650 mm/yr, runoff increased by 14 mm and sediment yield by 3.5 t ha^{-1} (Alicante, Spain). Values of post-fire sediment yields from the Western US reported by Moody and Martin (2009) were larger ranging between 240 t ha^{-1} for channels to 82 t ha^{-1} for hillslopes (with large variability across rainfall regimes). In other words, increased overland flow following wildfire enhances erosive intensity within channels where sediments accumulate in the pre-fire regime as mechanistically shown by Lamb et al. (2011). The increased potential for sediment mobilization combined with ash and disintegrated soil structure promote formation of high viscosity slurries (Burns and Gabet, 2015) that may induce catastrophic landslides (Thomas et al., 2021a) and debris flows in the first few years following a wildfire (McGuire et al., 2017).

Numerous studies have been devoted to quantifying wildfire effects on belowground soil functions (Neary et al., 1999) and impacts on hydrology (DeBano et al., 1998; Neary et al., 2005). The central and most intriguing question is how long the effects of wildfires on a landscape last? No simple and general answer is available as recovery rates vary with the severity of the wildfire impact, climate conditions, the property in question and vegetation dynamics. Ebel and Martin (2017) compiled information regarding rates of recovery of primary factors that control the landscape saturated hydraulic conductivity as an integrative soil parameter that controls infiltration into the soil. A summary of the results of Ebel and Martin (2017) is depicted in Fig. 26. Different landscape properties have different recovery time scales (Fig. 26a) that, in turn, affect the recovery of the landscape scale saturated hydraulic conductivity (Fig. 26b) and affect the partitioning of rainfall to infiltration and runoff. In agreement with established understanding of the critical roles of vegetation in soil structure formation and in shaping landscape scale saturated hydraulic conductivity and infiltration capacity (Bonetti et al., 2021). A recent study by Ma et al. (2020) demonstrated that landscape scale hydrologic recovery to pre-fire behavior is largely controlled by vegetation recovery as also reflected in measurable changes in evapotranspiration (ET). Availability of remotely sensed information on vegetation and spatial ET products can offers a means for estimating landscape hydrologic recovery though post-fire vegetation status.

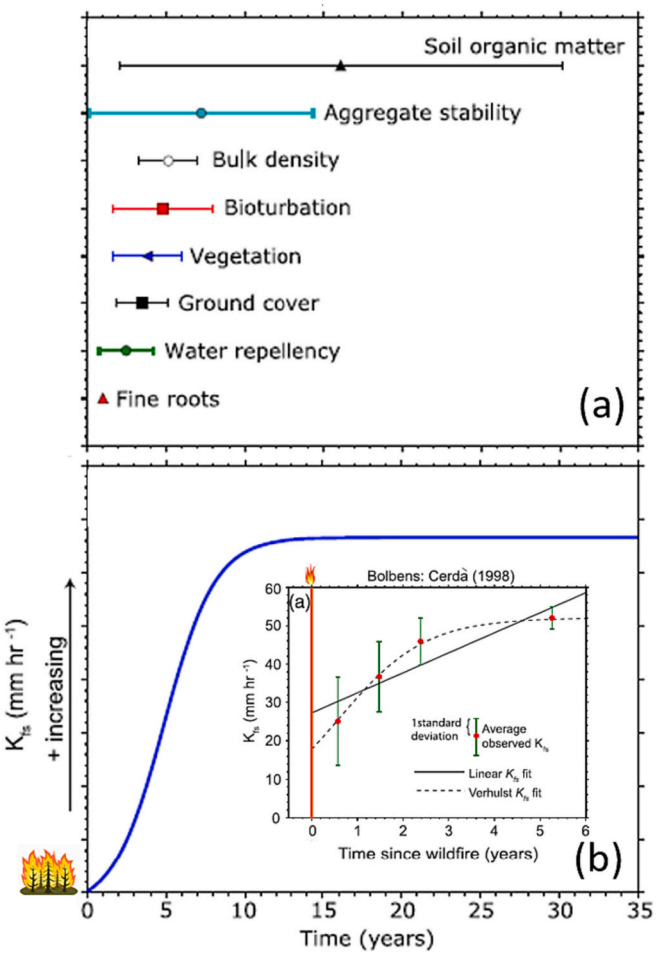


Fig. 26. (a) Time scales for post-fire recovery of factors affecting the saturated hydraulic conductivity at the landscape scale (y axis arbitrary for process separation); (b) Hypothetical recovery curve for saturated hydraulic conductivity with time (inset: measured saturated hydraulic conductivity values from a study in Spain) (adapted from Ebel and Martin, 2017).

10.5. Outlook and research gaps

Rapid advances in the field of wildland fire (wildfire) modeling offer unprecedented opportunities for quantitative translation of the various modeling capabilities to predictive assessment of wildfire effects on soil, hydrology and ecology of affected landscapes. The availability of common databases (soil, vegetation, hydrology, climate, fuel) that provide inputs to process-based models addressing environmental and wildfire science questions provide new opportunities for improving cross-disciplinary integration and information exchange. This review offers an entry point for non-wildfire scientists and modelers towards gaining insights into the wildfire science while identifying research needs and knowledge gaps in wildfire models as related to quantification of fire effects on soil, hydrology and ecology. While certain aspects of wildfire effects on soil and landscapes have been known for decades, including excellent studies addressing key concerns at the soil-wildfire interface reviewed here (Campbell et al., 1995; Albini et al., 1996; Enninful and Torvi, 2008; Massman, 2012; Massman, 2021), nevertheless, a coherent framework for systematically harnessing wildfire model outputs at landscape scales to quantify soil and hydrologic alterations (ecology and climate feedbacks implicitly included here) is lacking. Among the many research questions that can be addressed by a framework like the one proposed here, we list a few important ones such as:

- o How do soil properties change with wildfire characteristics? And how best to quantify and scale spatially-varying fire intensity effects on hydrologic properties?
- o How do wildfires impact soil at scales from plots (where measurements are often made) to watershed scales where they matter most?
- o How impacts of wildfires on portions of a basin can be integrated to predictions of hydrologic-climatic responses at basin outputs (e.g., stream flows, nutrient fluxes, sediment yields)?
- o How post-fire soil and landscapes evolve with time along different recovery trajectories and what mechanisms (and rates) restore different functions?
- o How can different models be used to advantage in predicting wildfire impacts and in guiding response and restoration activities? And what are the data sets necessary for verifying the accuracy of wildfire models?
- o How to engage stakeholders in selection of desired landscape recovery outcomes?

Landscape management and climate jointly affect the frequency and size of wildfires and determine the persistence of post-fire effects as well as the response of fire-affected landscapes and their recovery rates. Anecdotal evidence points to significant alterations of regional water supply and quality of fire affected basins (Williams et al., 2022; Ma et al., 2020; Ebel et al., 2023; Whelton et al., 2023) and rates of forest regeneration under increasingly warm and dry climate (Niemeyer et al., 2020; Stewart et al., 2021; Davis et al., 2023). While all actors and compartments of fire-affected landscapes affect the ecohydrological response to fire disturbances, soil alterations play a critical role in all aspects of the response and recovery, yet soil representation in these processes remain largely qualitative. We advocate for injection of quantitative representation of how fire-induced soil alterations affect regional hydrology and what are the (quantitative) rates and pathways for soil function and trait recovery.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Sean McKenna reports financial support was provided by Desert Research Institute.

Data availability

Data will be made available on request.

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